

8

7

6

5

4

3

2

1

AFFINITY

SYSTEM 4.0
04-S28-2014
02/09/15

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED

MULTIMEDIA

AP_PMIC

RF INTF

BLK_WTR2955

B_RFFE

ANTENNAS

AUDIO_FE

XMCS

CONNECTIVITY

POWER

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA MOBILITY		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri Dec 18 13:27:05 2015		84016530001_a	
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CAD FILE NAME:		PAGE 1 OF 52	

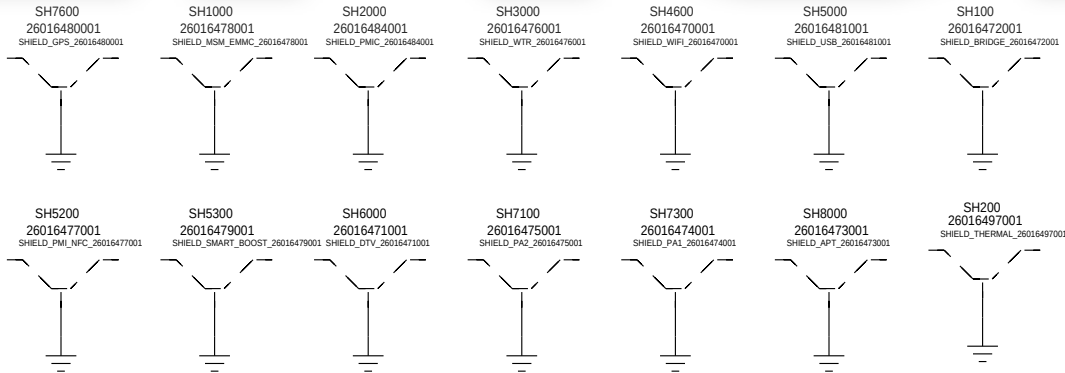
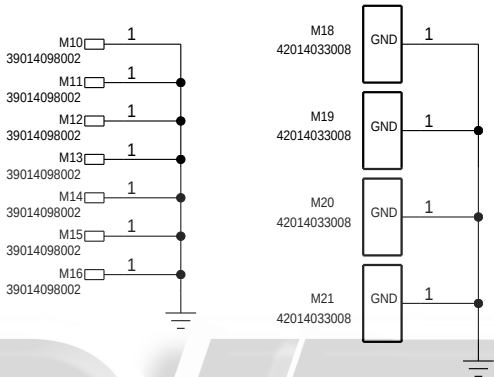
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

AFFINITY ME/SHIELDS

AFFINITY REFERENCE DESIGNATOR RANGES BY BLOCK

- 000 - 499: ANTENNA
- 500 - 599: USER INPUT
- 600 - 799: POWER
- 1000-2999: MSM8852 + PM8952
- 3000-3049, 3100-3149: WTR2955
- 3050-3099, 3150-3199: CA_WTR2955
- 3200-3999: MSM8952 + PM8952
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO FE
- 5000-5499: DISPLAY
- 5500-5999: CAMERA
- 6000-6999: XMCS
- 7000-8999: RFFE
- 9000-9999: PRODUCT-SPECIFIC (FILTERING,, 100PF CAPS)

UNIVERSAL GROUND CONTACTS



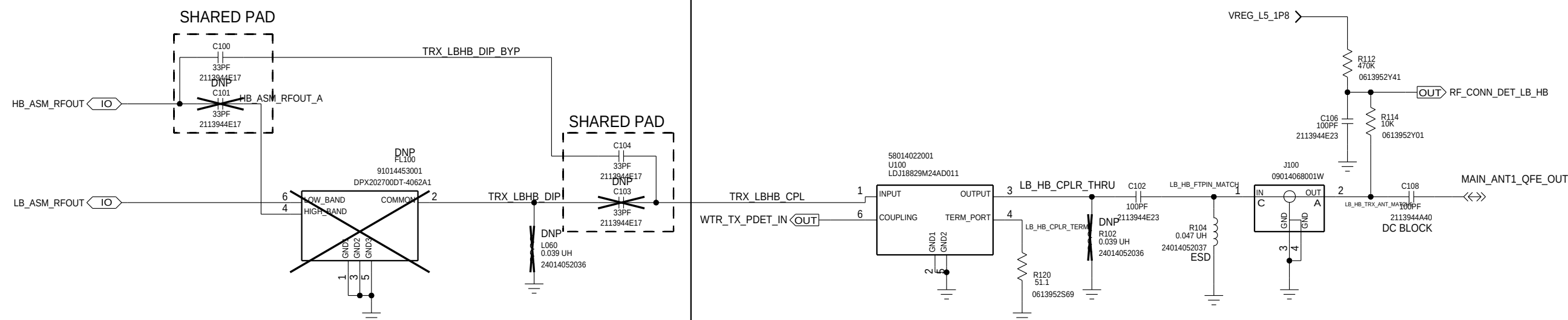
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Fri Feb 19 14:26:03 2016		84016530001_a	
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		REV 01 PAGE 2 OF 52	

REF:100-114

DEVICE LOCATION - BOTTOM RIGHT

MAIN LOW / HIGH BAND ANTENNA (GSM/UTMS/LTE)

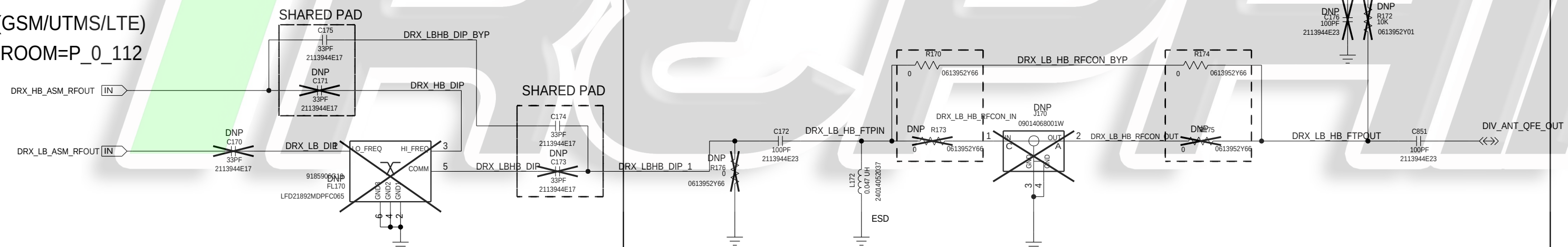


REF:170-179

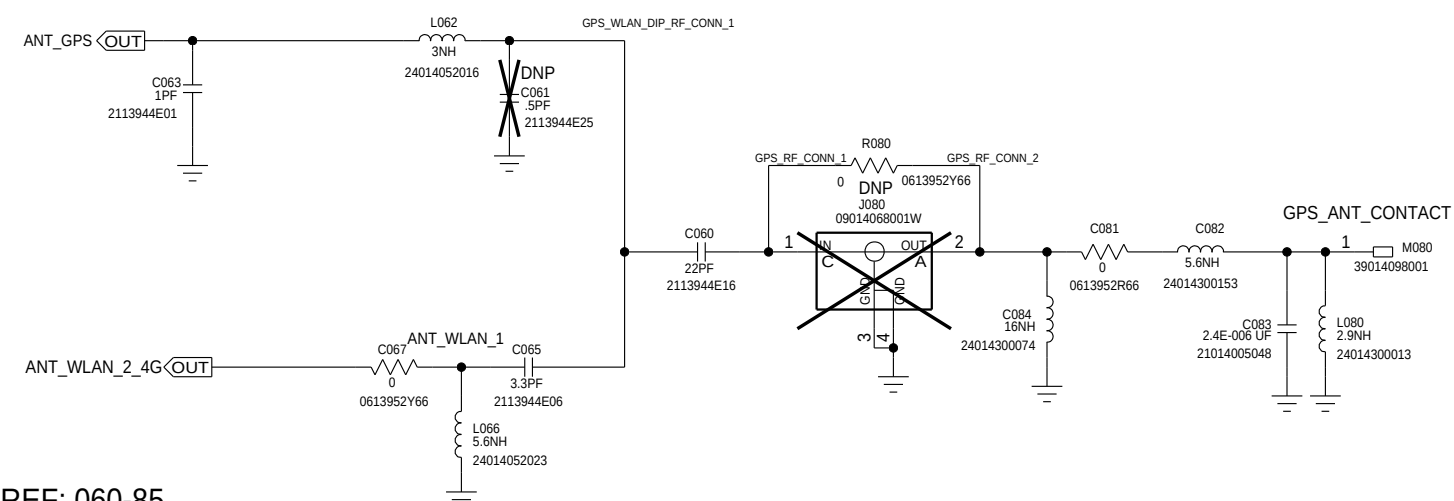
DEVICE LOCATION - TOP LEFT

(GSM/UTMS/LTE)

ROOM=P 0 112

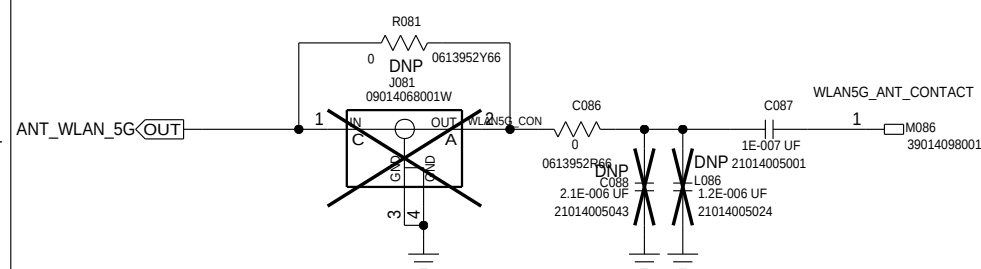


GPS/BT/WLAN ANTENNA
SIDE ANTENNA (ABOVE VOLUME KEYS)



WLAN 5G ANTENNA

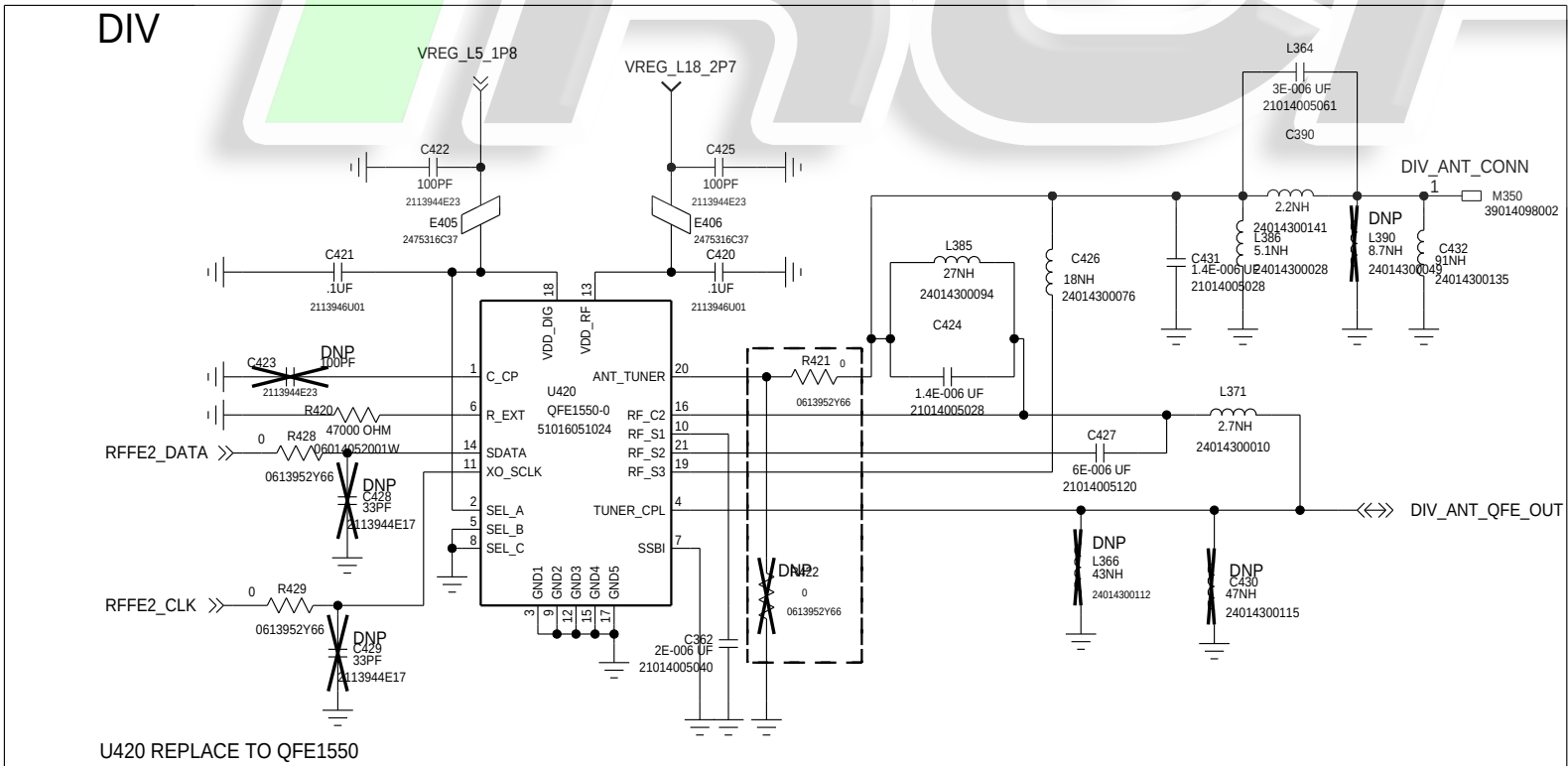
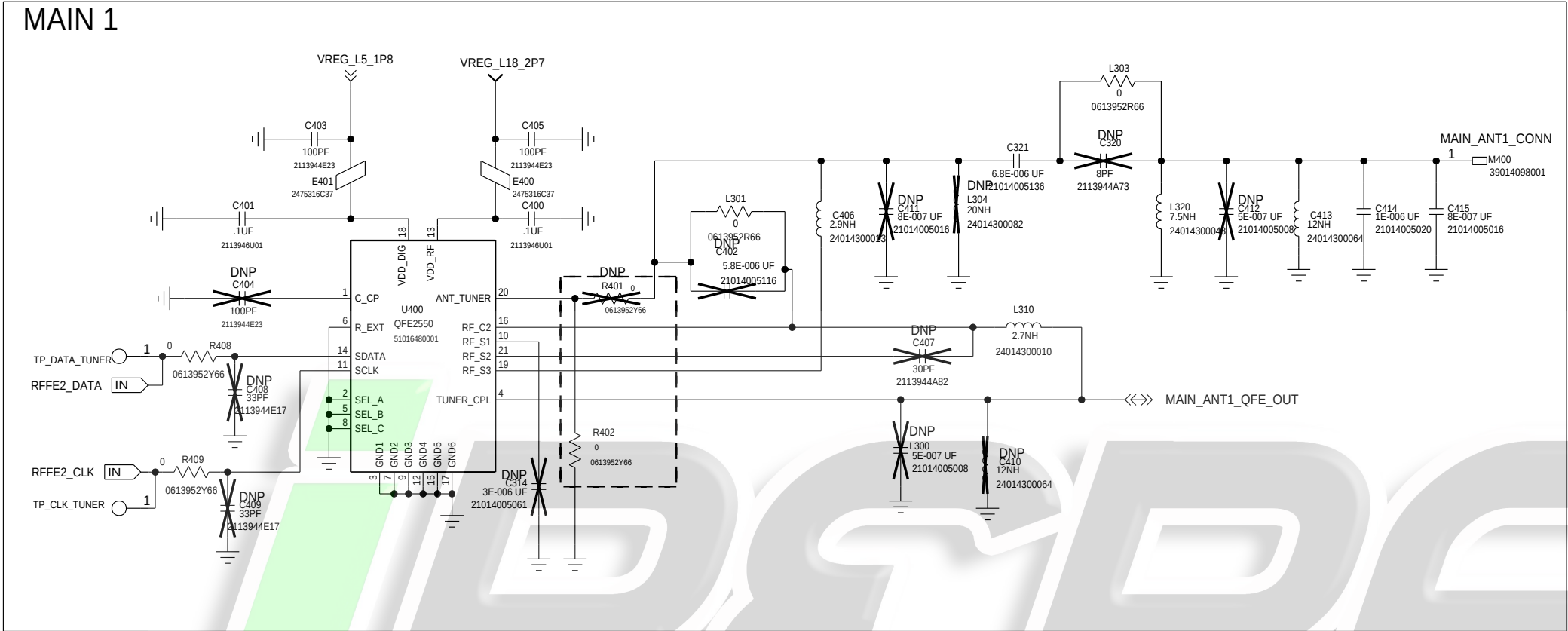
SIDE ANTENNA (ABOVE VOLUME KEYS)



APPROVALS		MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: antennas		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:04 2016				
MOTOROLA CONFIDENTIAL RESTRICTED		C	PROJECT NAME 84016530001_a.cpm	REV 01
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ANT: QCOM TUNERS

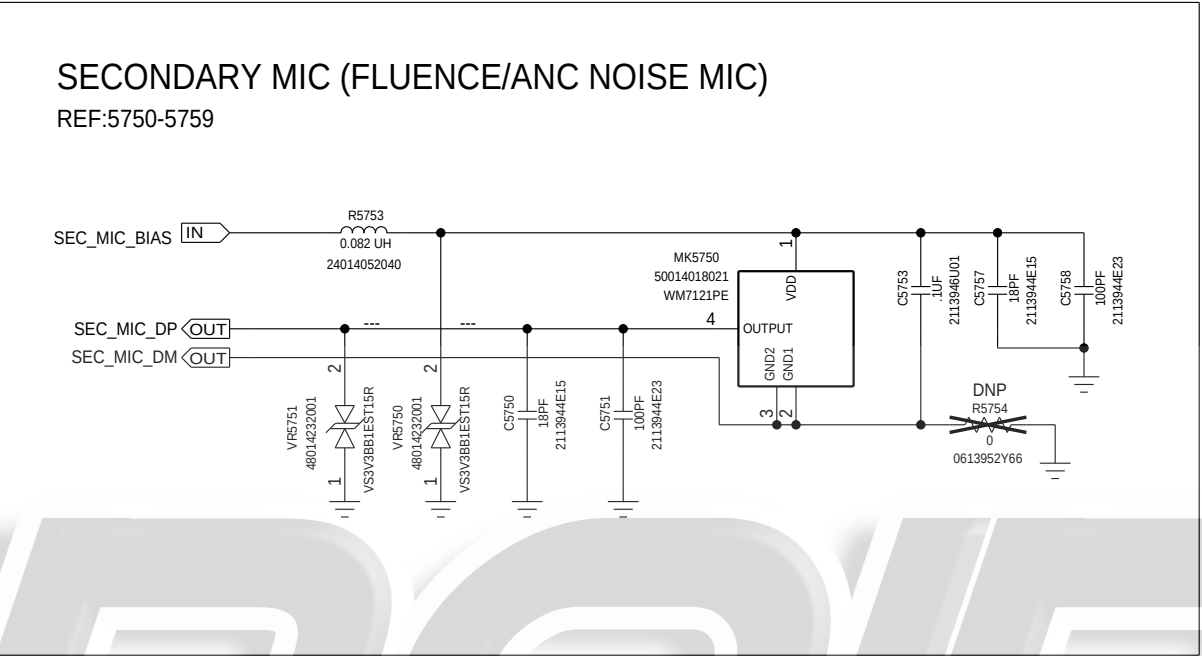
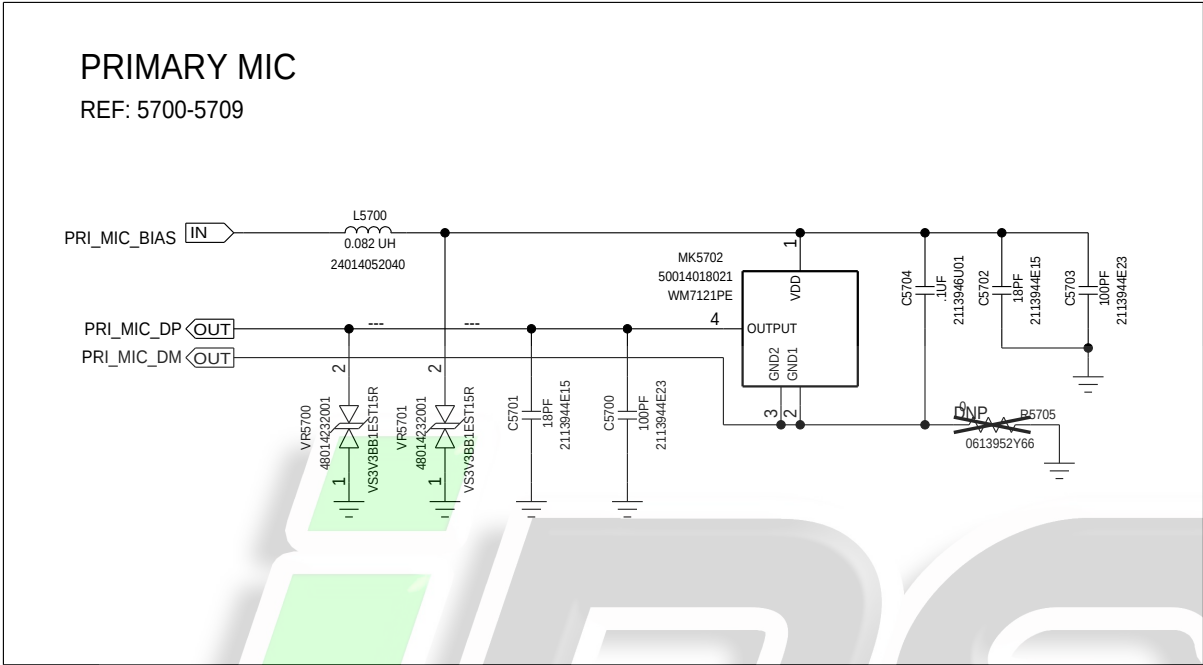
REFDES: 400-499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: QIUMING LI Fri Feb 19 14:26:05 2016		antennas	
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		CAD FILE NAME:	
		PAGE 4 OF 52	

AFE: MICS

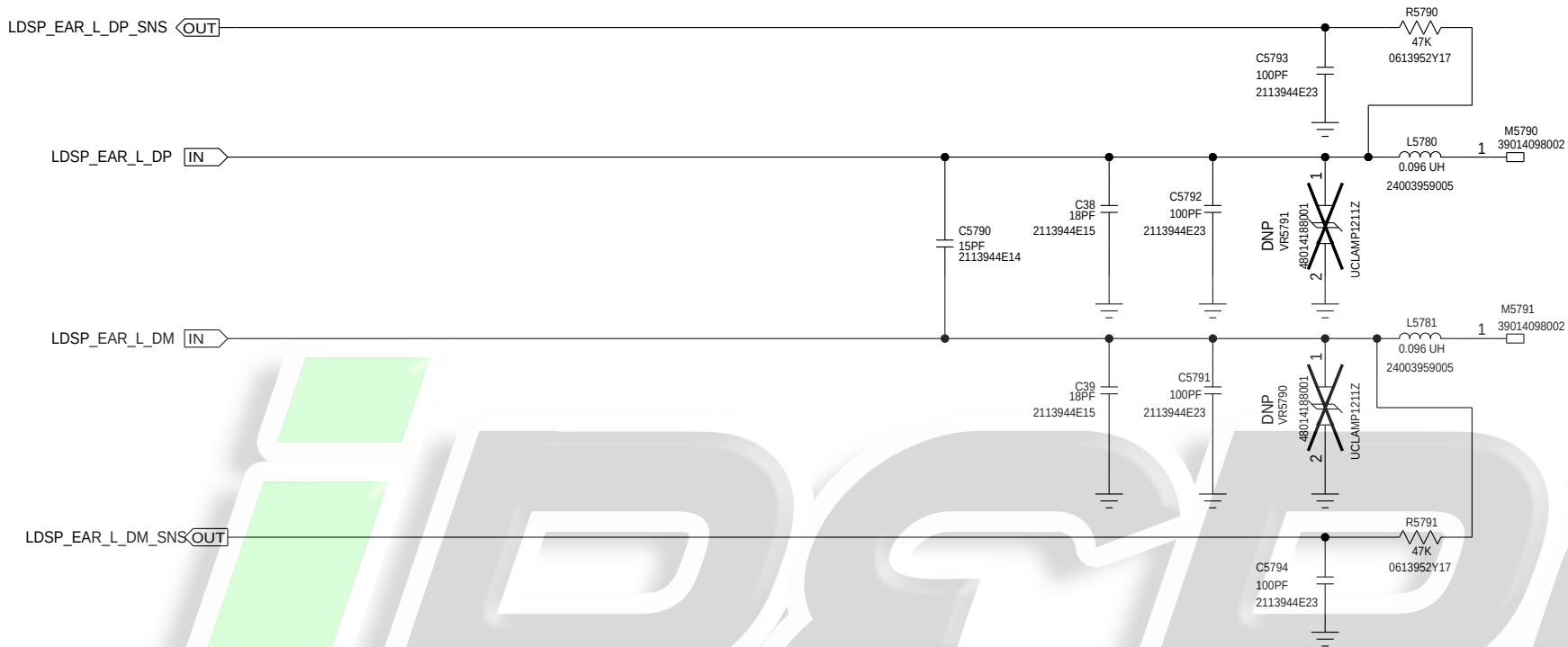
REF: 5700-5799



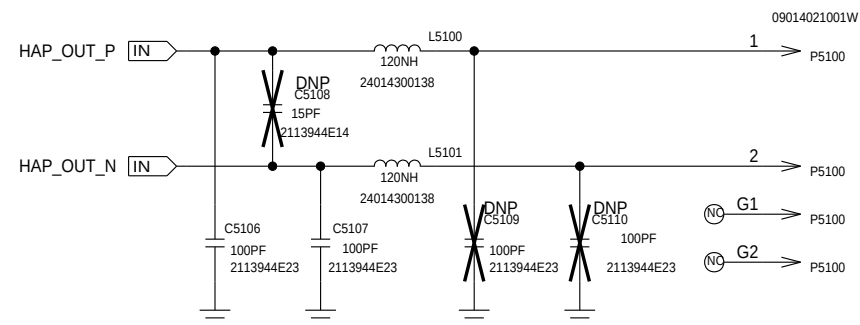
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: DENNIS BUDNICK		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri Feb 19 14:26:06 2016		audio_fe	
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		CAD FILE NAME:	
		REV 01	
		PAGE 5 OF 52	

AFE: SPEAKERS

LOUDSPEARKER/EAR PIECE
REF:5790-5799



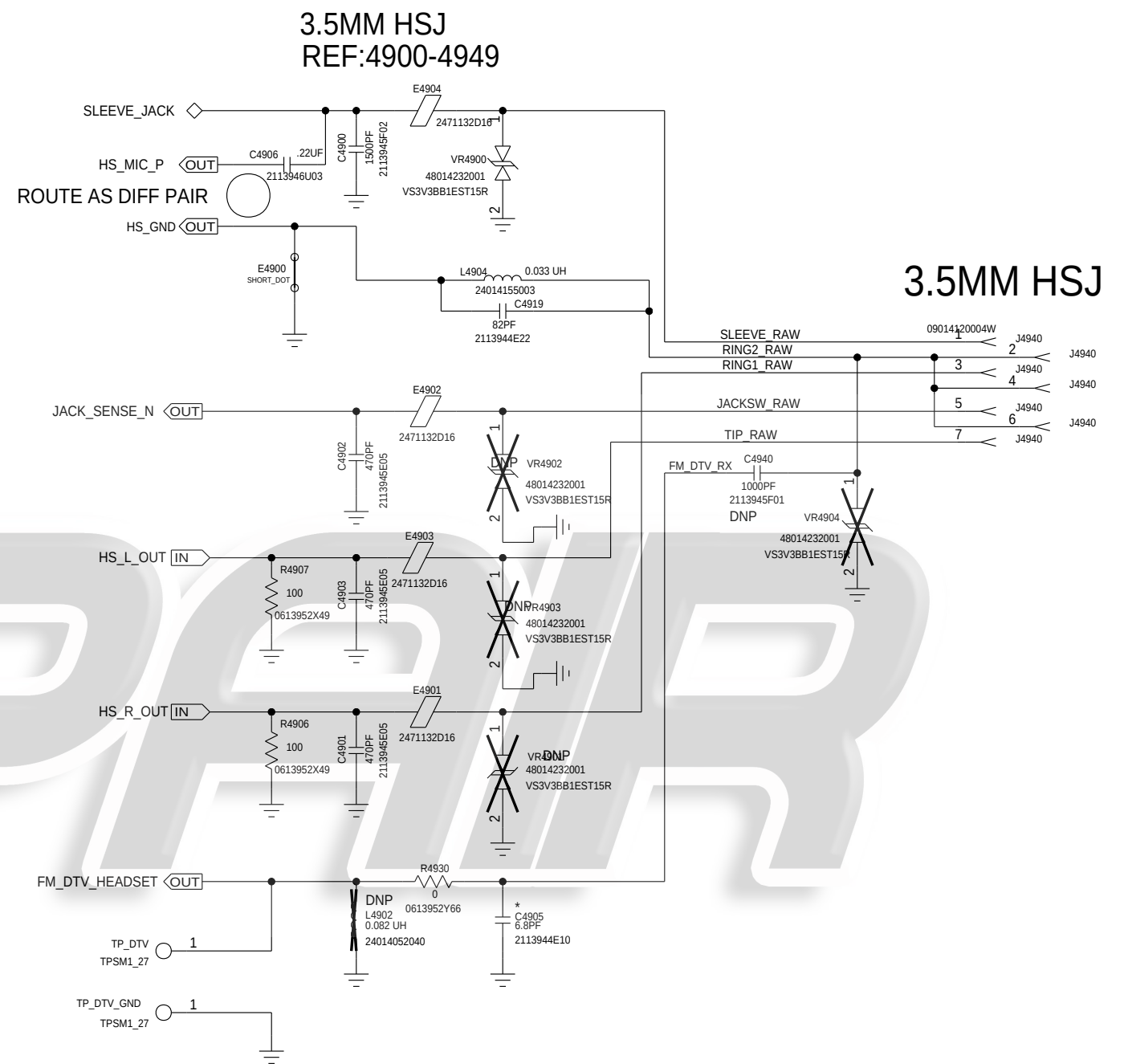
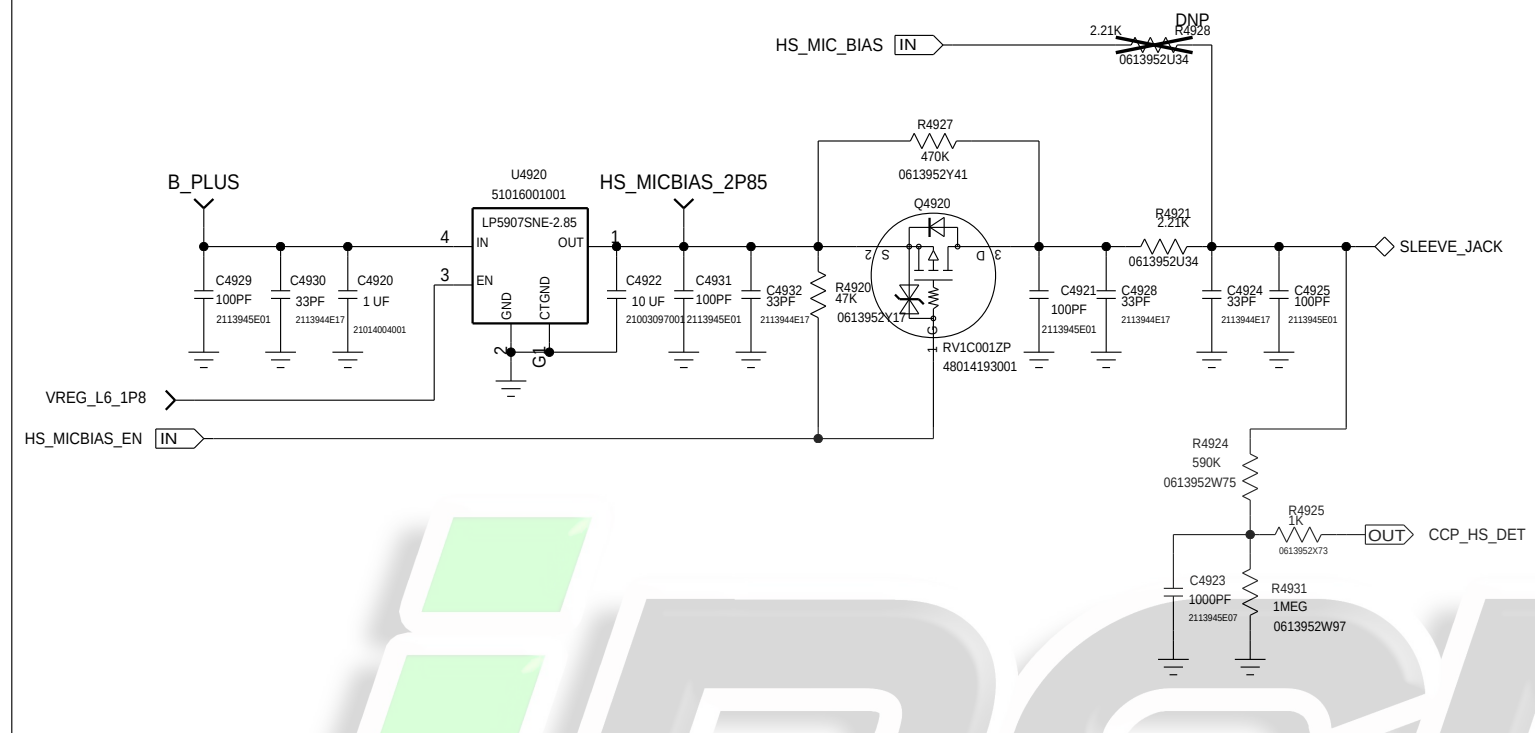
ROTARY VIBRATOR
REF 5100-5119



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: DENNIS BUDNICK		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: TIM MILLER Fri Feb 19 14:26:08 2016		PROJECT NAME 84016530001_a.cpm	
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		PAGE 6 OF 52	

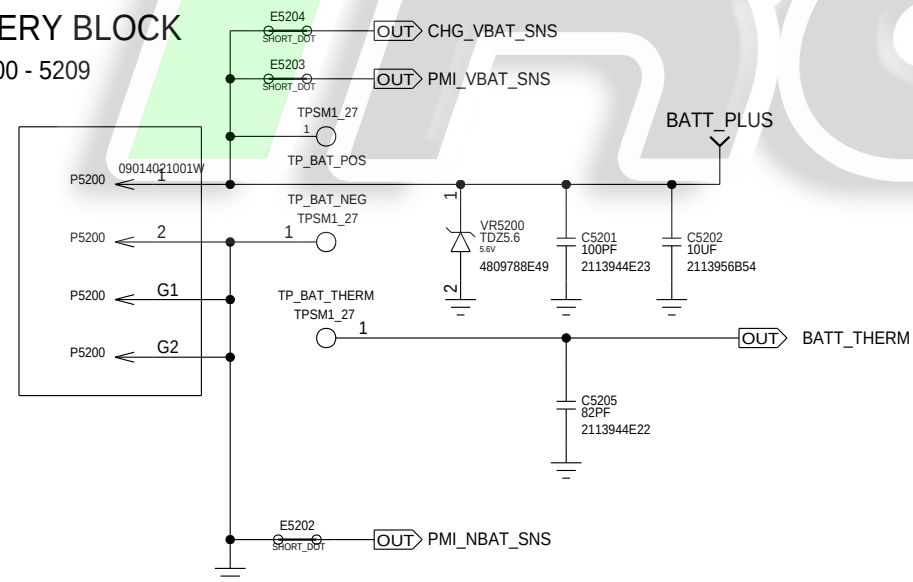
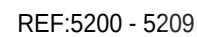
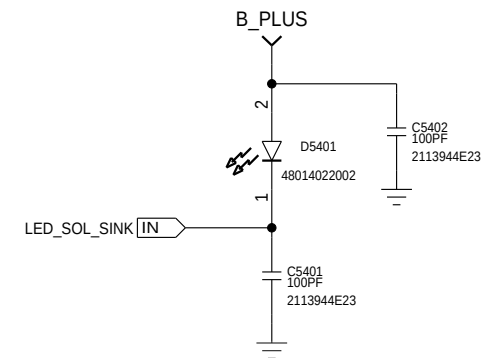
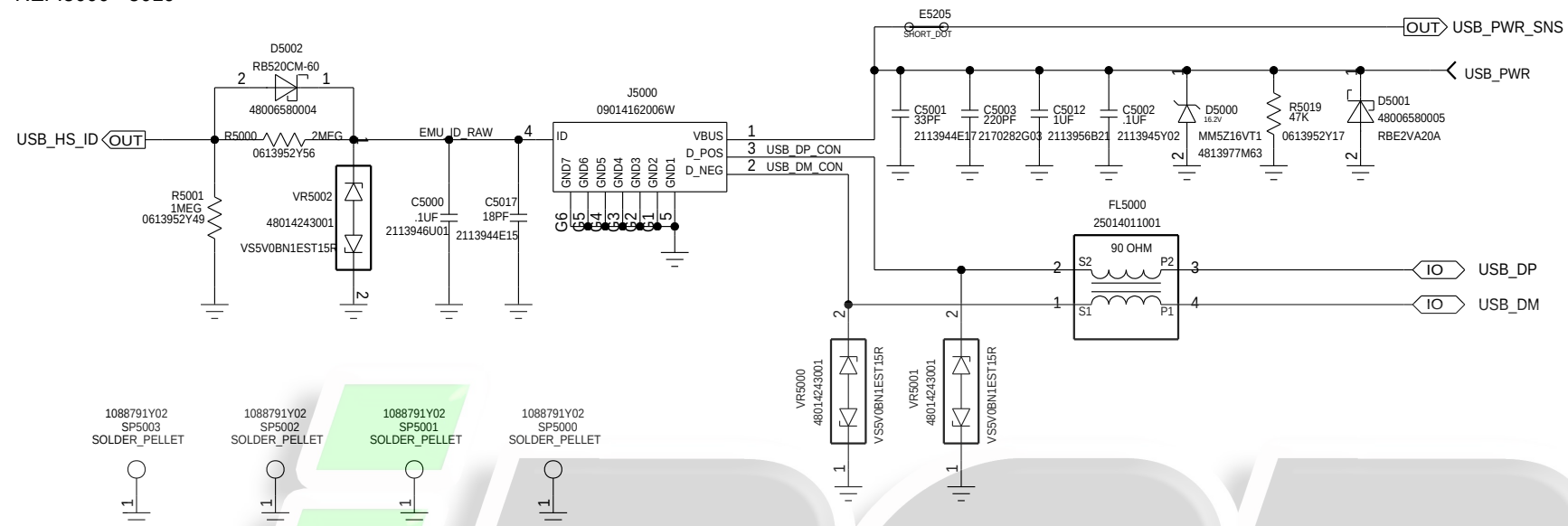
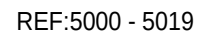
AFE: 3.5MM HEADSET

SENSOR HUB HEADSET DETECTION CIRCUIT



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:07 2016		PROJECT NAME 84016530001_a.cpm	
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		PAGE 7 OF 52	

REF:5000 - 5219



USING MAXIM FOR TEMP SENSE: DNI R5206, R5204 & R5208 PLACE R5209
USING PMIC FOR TEMP SENSE: PLACE R5206, R5204 & R5208 DNI R5209

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG	BLOCK TITLE: power		
LAST MODIFIED BY: NATE CONNELL Fri Feb 19 14:26:09 2016			
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		CAD FILE NAME:	PAGE 8 OF 52

D

C



A

10

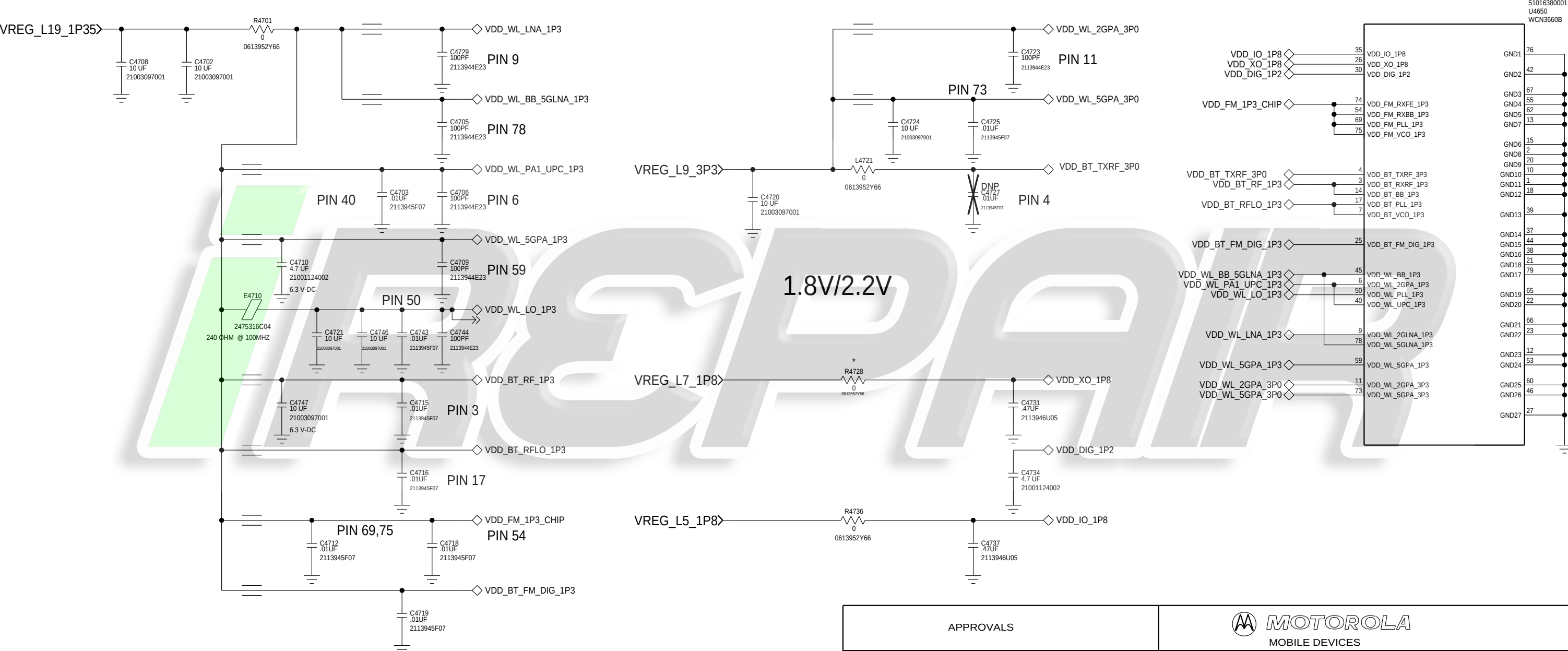
A

WCN3660B POWER
REF 4700-4799

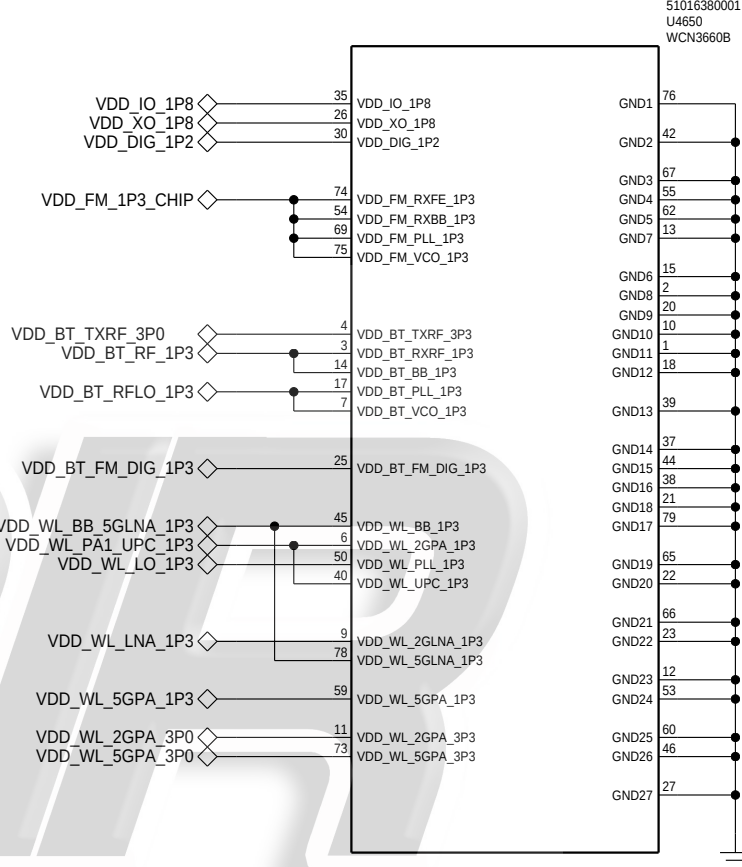
NEED TO SCRUB ALL SUPPLY SEQUENCE/VOLTAGE/CURRENT NUMBERS FOR WCN3660B

BT_FM_WLAN_1P3

WLBT_3.0V



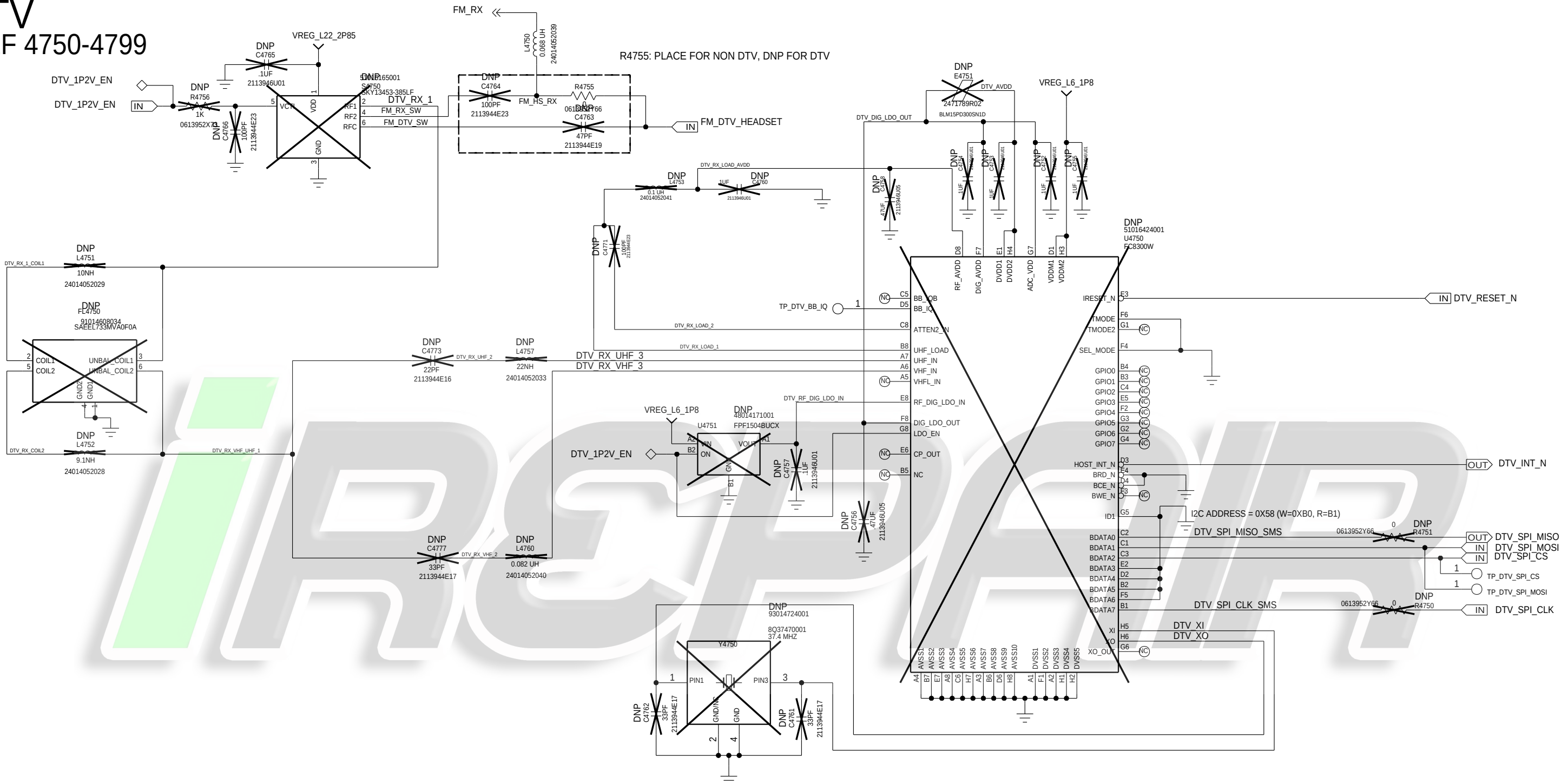
NOTE: RESERVE C4717 AND C4718 FOR 0.01UF BYPASSES



BASED ON 80-WL006-44 REV C

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Fri Feb 19 14:26:10 2016		connectivity	
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		CAD FILE NAME:	PAGE 10 OF 52

DTV
REF 4750-4799



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: GSHIMON Fri Feb 19 14:26:11 2016		connectivity	
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		CAD FILE NAME:	PAGE 11 OF 52

CONN: UIM 1& 2

UIM1 CONN
REF:5500-5509

UIM_PRESENT:
HIGH WHEN CARD PRESENT (OPEN)
LOW WHEN CARD ABSENT (CLOSED)

VREG L5_1P8 VREG L14_UIM1

UIM1_POWER VCC_SIM1

UIM1_SWP <<=>

UIM1_DATA IO

UIM1_RESET IN

UIM1_CLK IN

UIM1_PRESENT OUT

UIM2 CONN
REF:5520-5529

UIM_PRESENT:
HIGH WHEN CARD PRESENT (OPEN)
LOW WHEN CARD ABSENT (CLOSED)

VREG L5_1P8 VREG L15_UIM2

UIM2_POWER VCC_SIM2

UIM2_SWP <<=>

UIM2_DATA IO

UIM2_RESET IN

UIM2_CLK IN

UIM2_PRESENT OUT

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Fri Feb 19 14:26:11 2016		connectivity	
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		CAD FILE NAME:	PAGE 12 OF 52

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

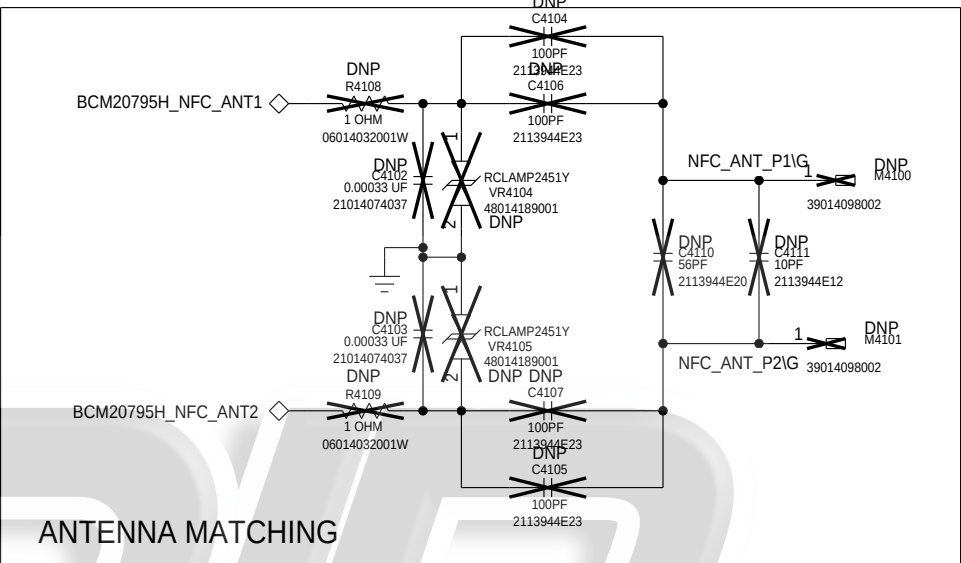
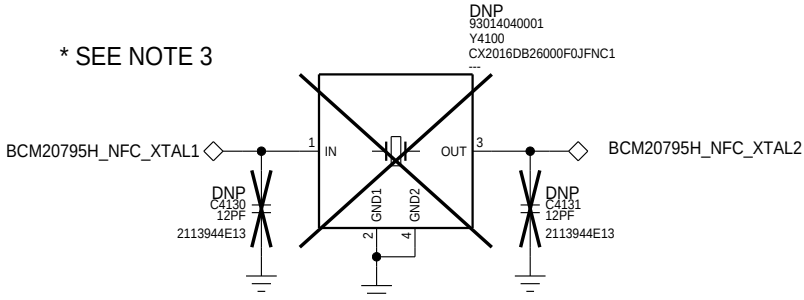
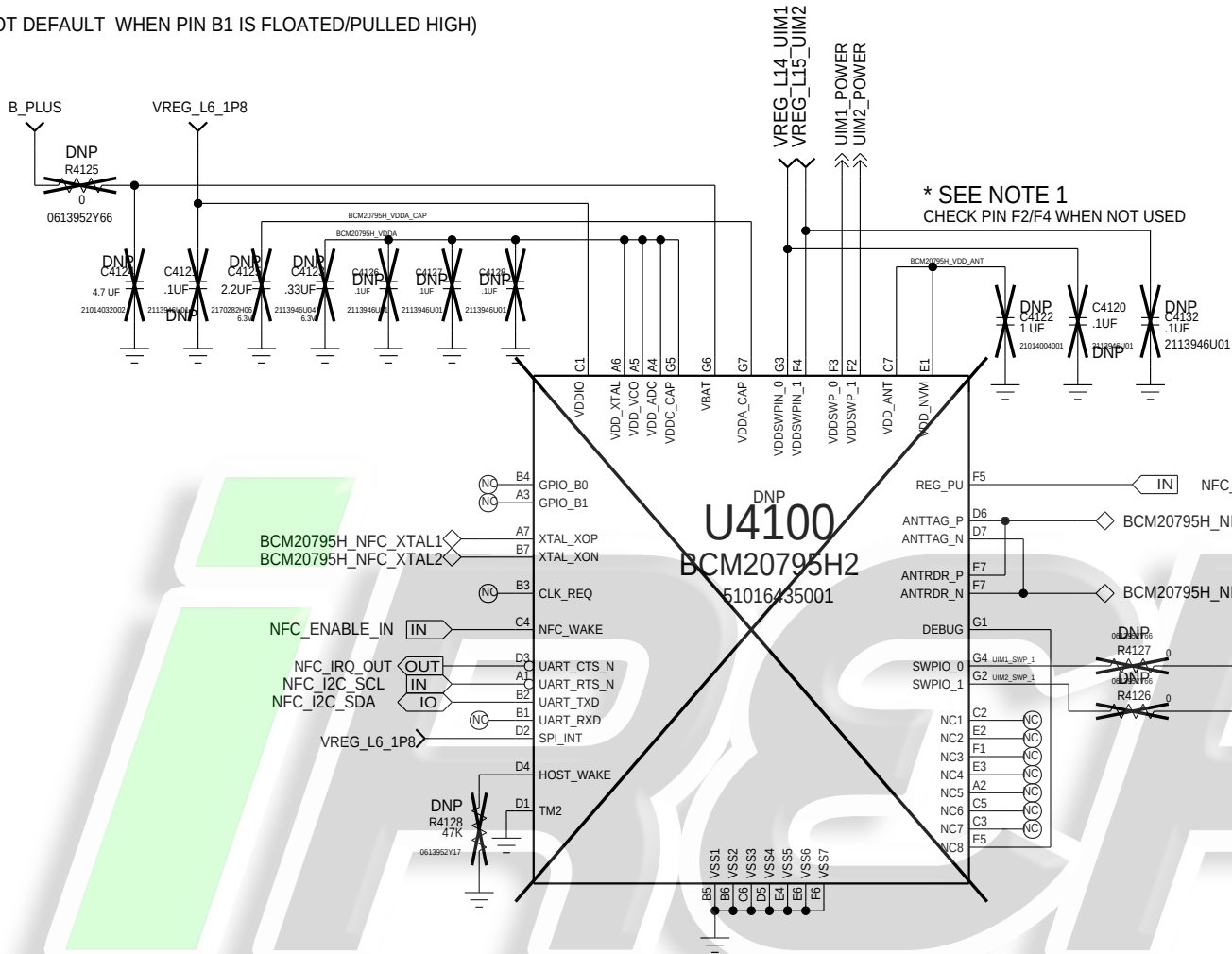
BROADCOM NFC

REF: 4100-4199

I2C ADDR: 0X77 (BOOT DEFAULT WHEN PIN B1 IS PULLED TO GROUND)

OR

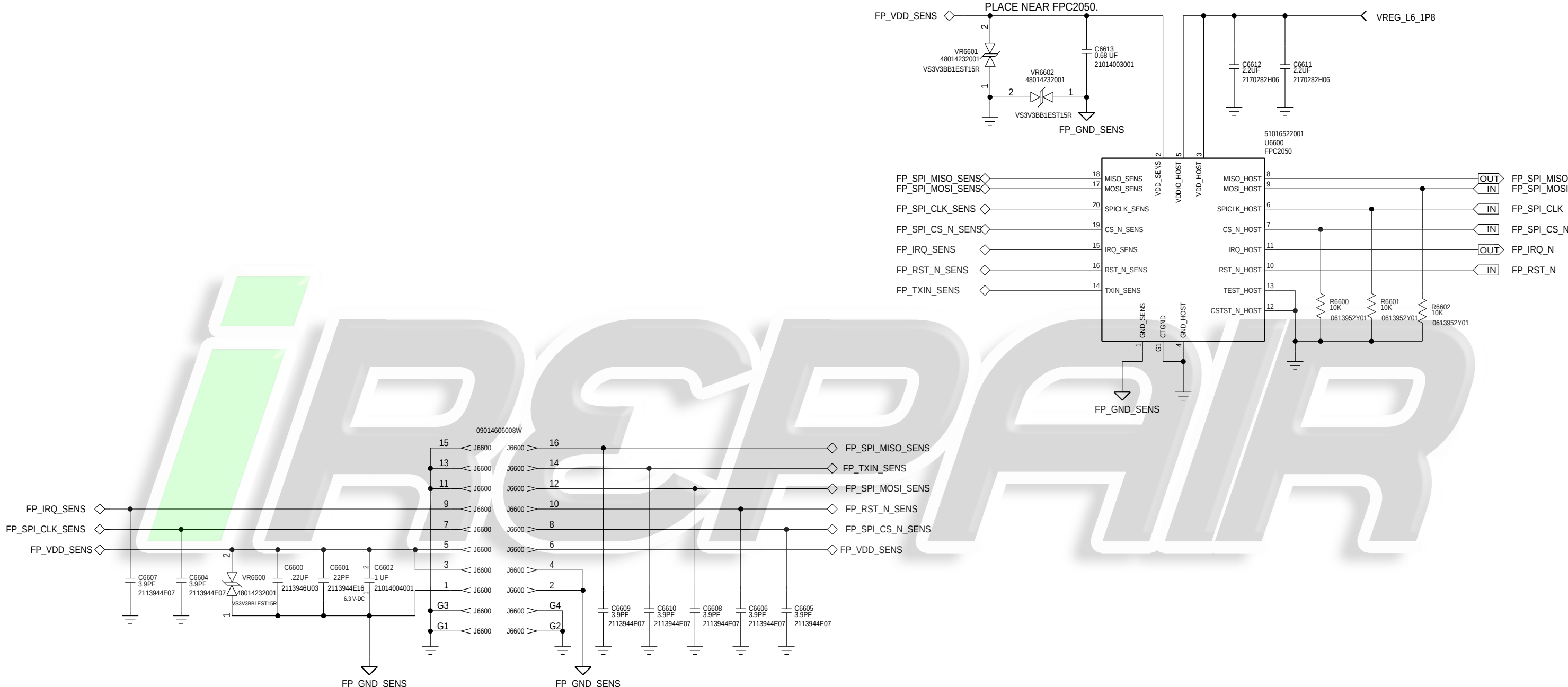
I2C ADDR: 0X76 (BOOT DEFAULT WHEN PIN B1 IS FLOATED/PULLED HIGH)



- NOTES:
1. PER BROADCOM RECOMMENDATION, CAPACITANCE ON UIM1_POWER NET SHOULD BE 0.33UF. TOO LITTLE COMPROMISES REGULATOR STABILITY, TOO MUCH CAN CAUSE INADVERTENT POWER-ON RESETS AT VDD.
 2. PLACE U4101 AND C4100 ONLY ON CHINA SKU
 3. FOR SMALL NFC ANTENNAS, EXTERNAL CRYSTAL OSCILLATOR CIRCUIT MUST BE USED - SEE EE8994CORE-296 FOR DETAILS. IF USE OF EXTERNAL CLOCK IS DESIRED, CHECK WITH BROADCOM BEFORE COMMITTING.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Fri Feb 19 14:26:14 2016		connectivity	
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		REV 01	
		CAD FILE NAME:	
		PAGE 13 OF 52	

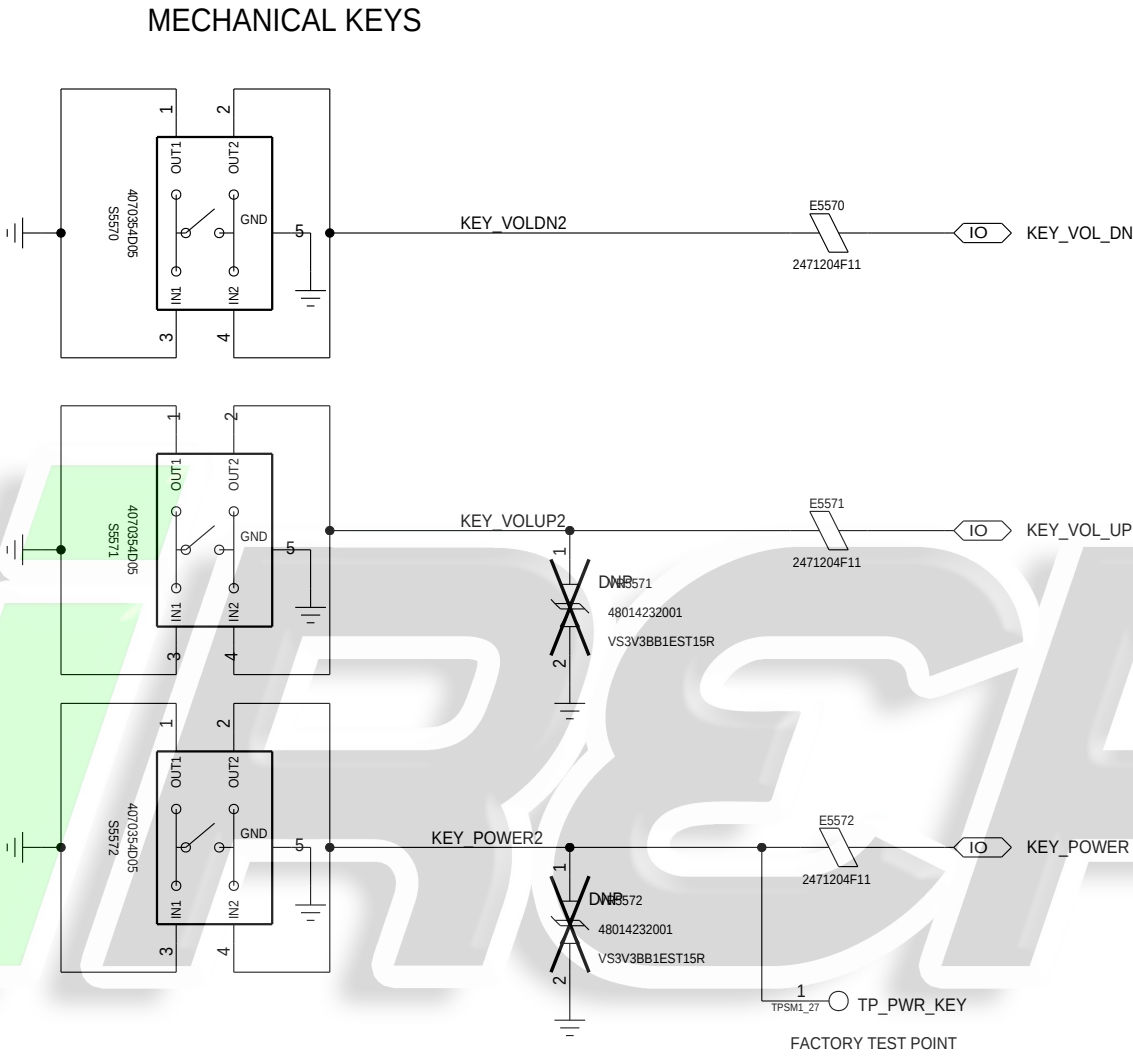
USR: FPC



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: connectivity	
LAST MODIFIED BY: NATE CONNELL Fri Feb 19 14:26:15 2016			
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		CAD FILE NAME:	PAGE 14 OF 52

SIDEKEY SWT

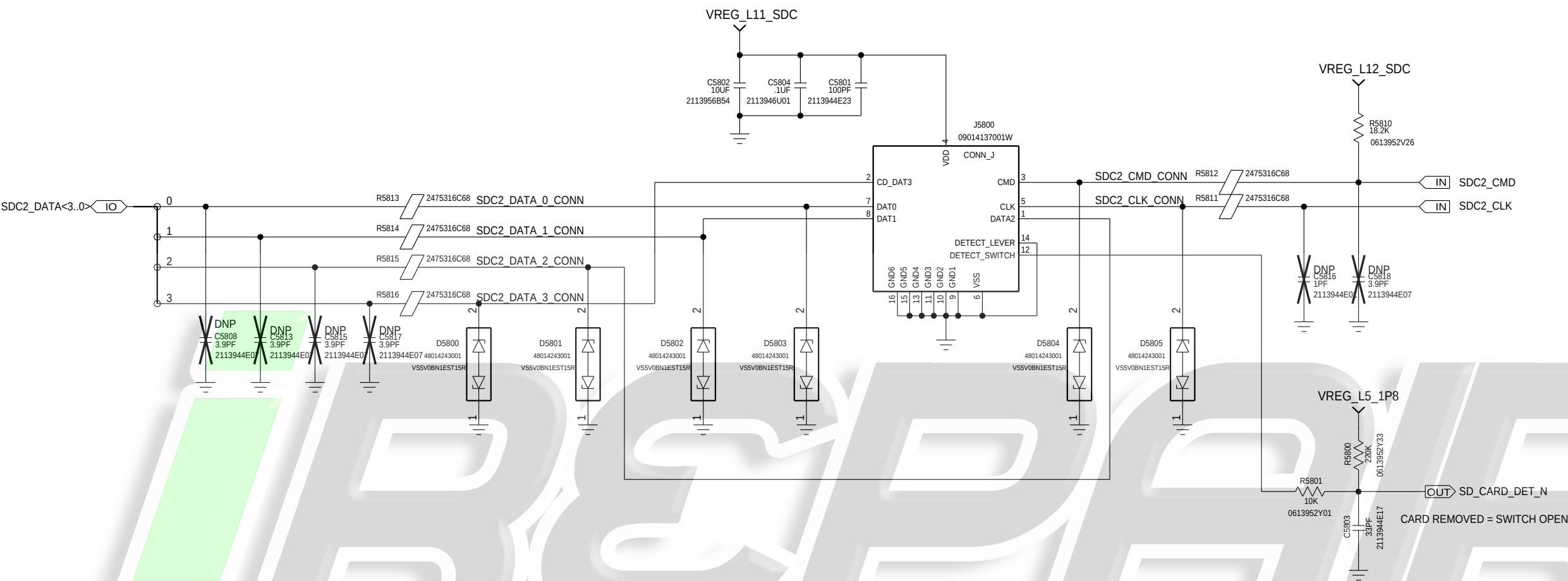
REF:5570-5579



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri Feb 19 14:26:14 2016		connectivity	
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		CAD FILE NAME:	
		PAGE 15 OF 52	

USR: MICRO SD

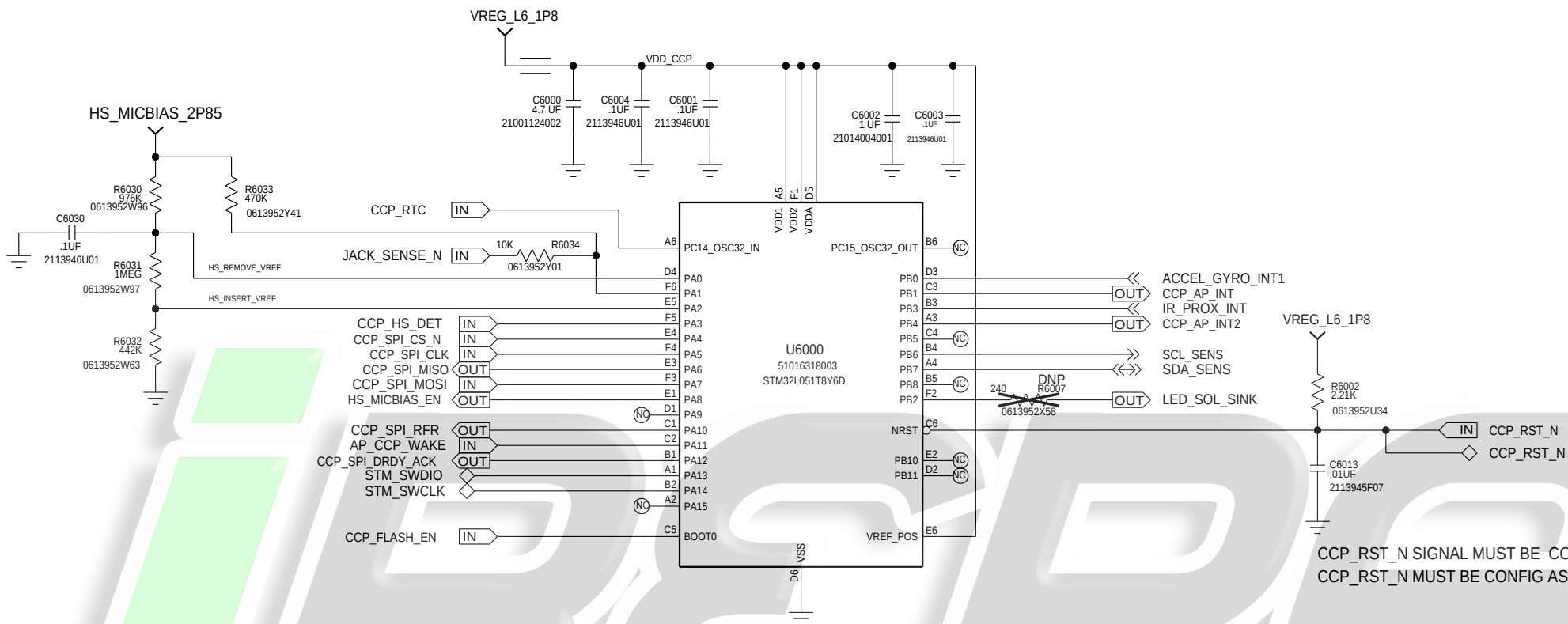
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:13 2016		connectivity	
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		CAD FILE NAME:	
		PAGE 16 OF 52	

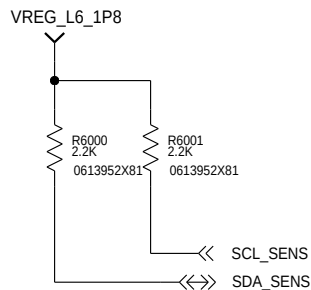
XMCS: CCP

REF: 6000-6049

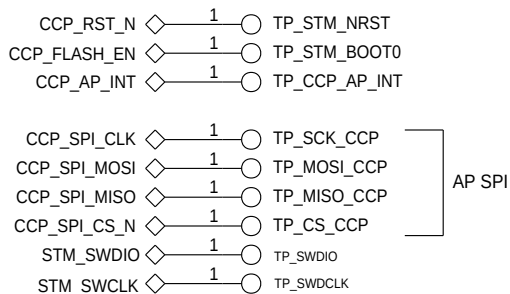


CCP_RST_N SIGNAL MUST BE CONFIG AS OUTPUT LOW FROM MSM TO RESET CCP
CCP_RST_N MUST BE CONFIG AS INPUT TO MONITOR RESET OUTPUT FROM CCP

I2C SUPPORT



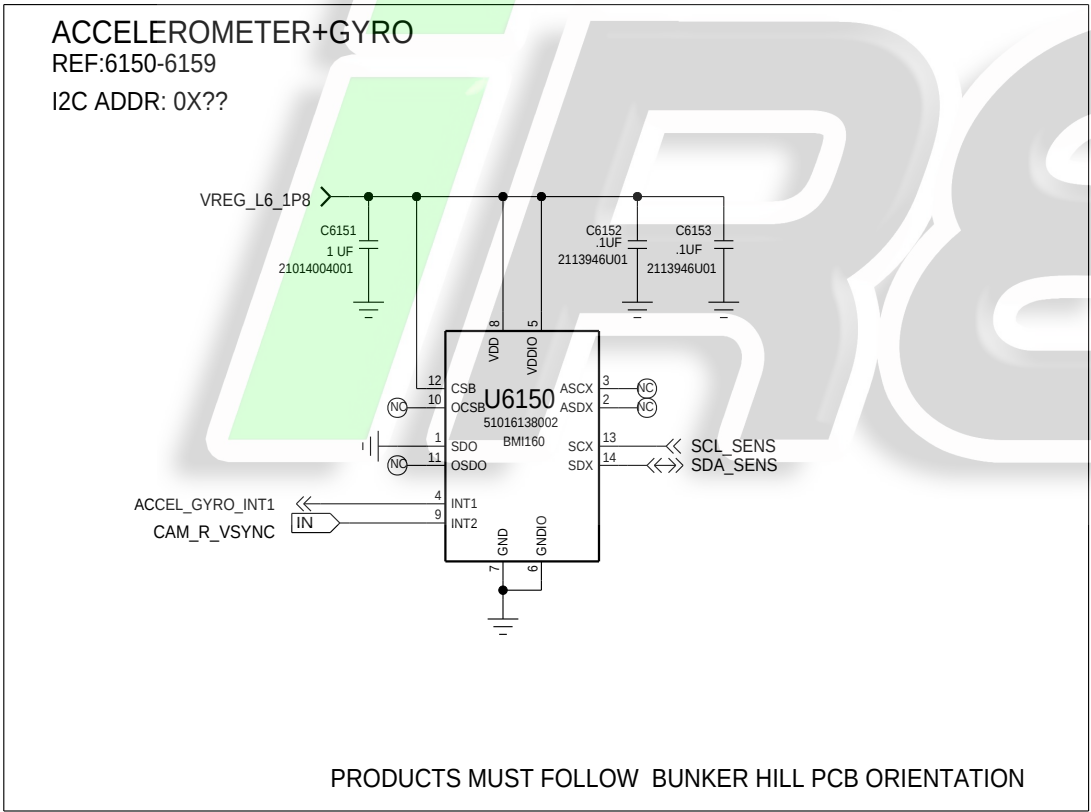
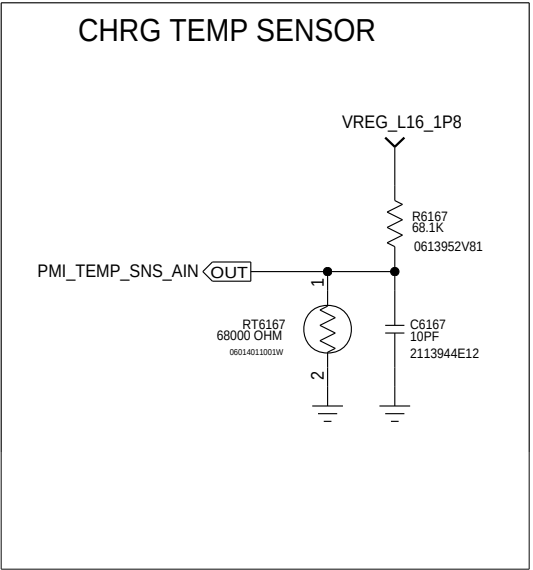
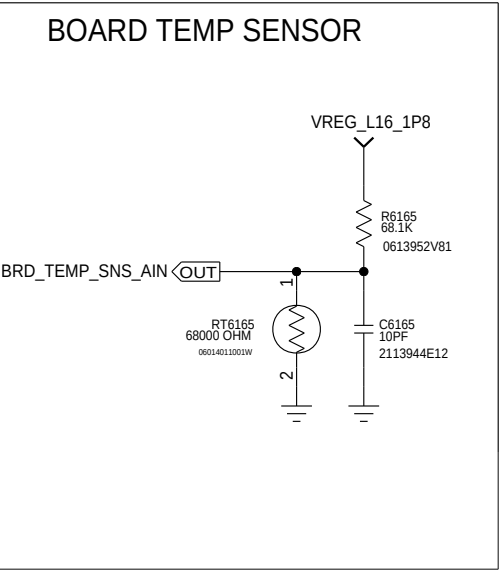
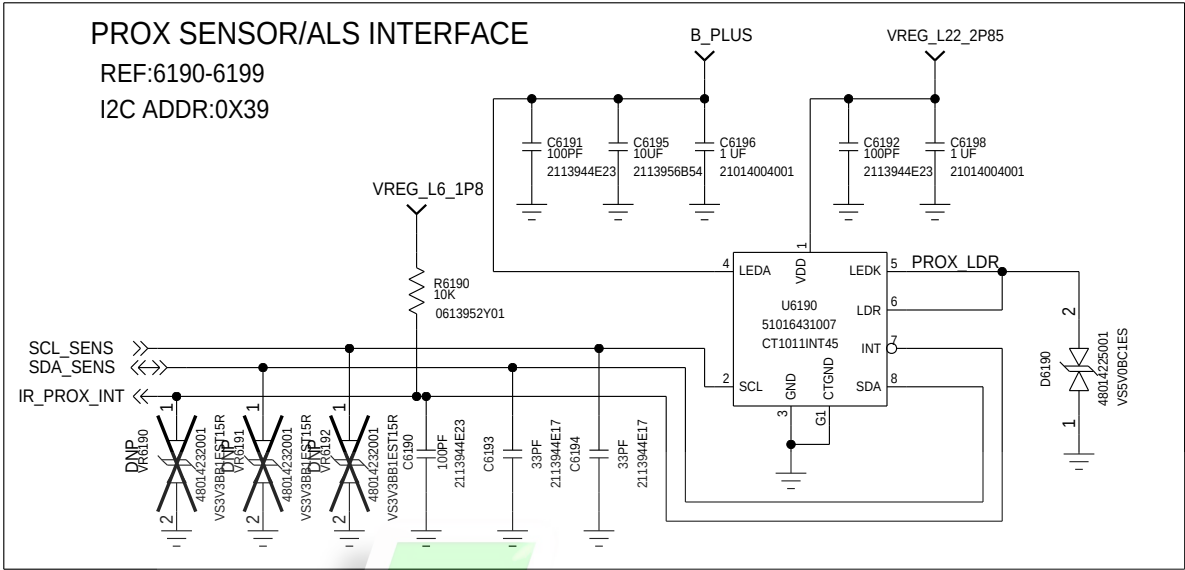
TESTPOINTS



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN GORSICA		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri Feb 19 14:26:16 2016		xmcs	
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		PAGE 17 OF 52	

XMCS: SENSORS

REF: 6100-6199

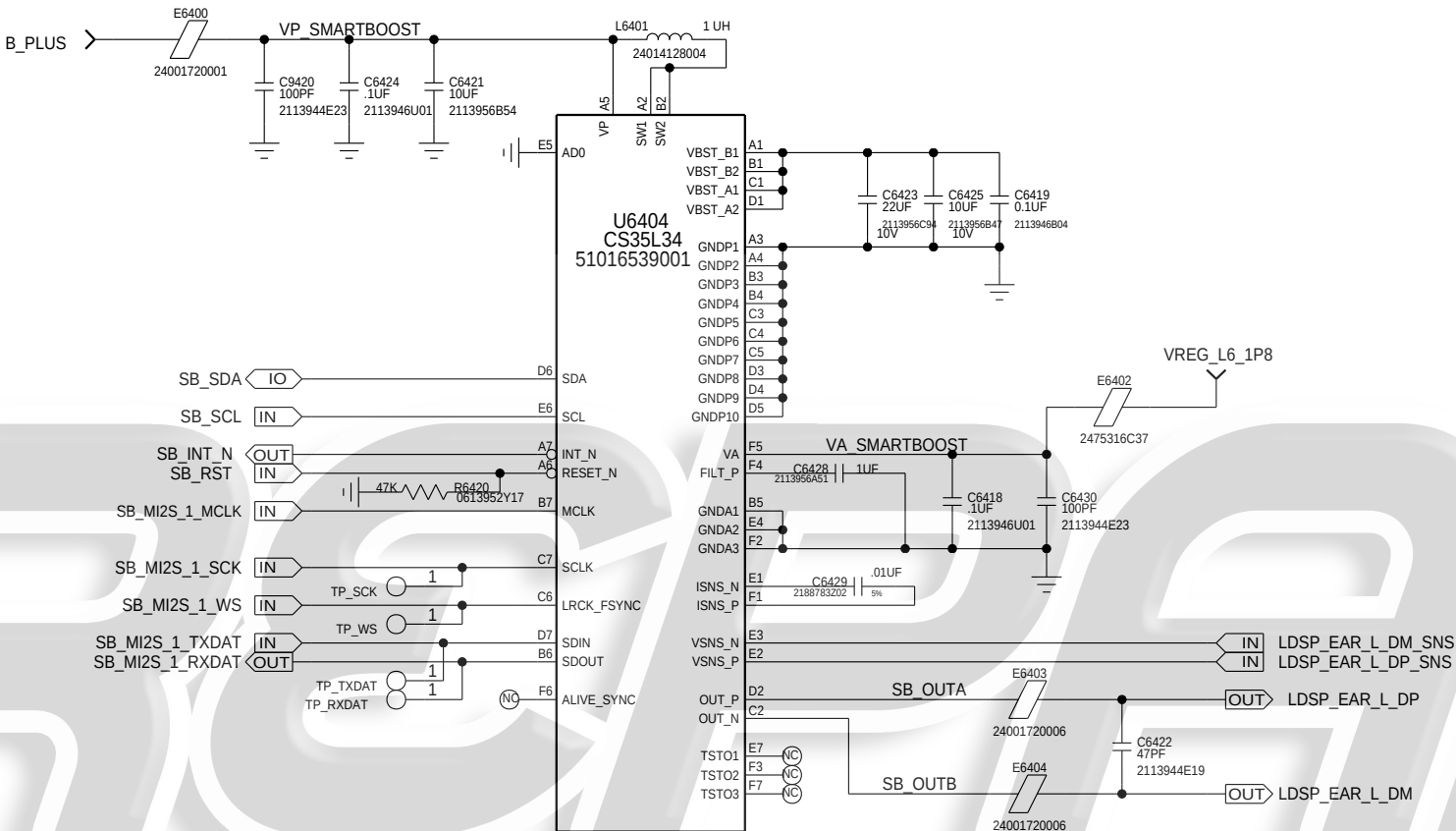


PRODUCTS MUST FOLLOW BUNKER HILL PCB ORIENTATION

JOHN GORSICA/NATE CONNELL APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: HUNG PHAM		BLOCK TITLE:	
LAST MODIFIED BY: HUNG PHAM Fri Feb 19 14:26:17 2016		xmcs	
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		B 84016530001_a.cpm	
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XMCS: SMART BOOST WATTSON

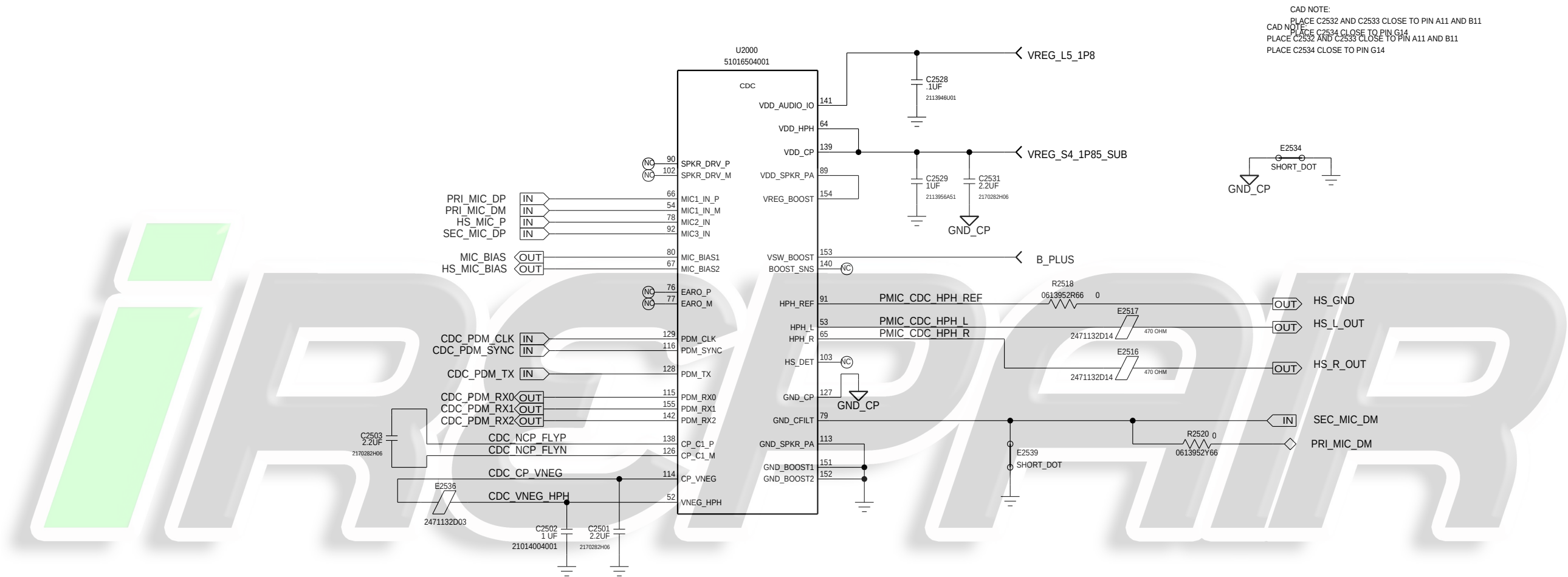
REF 6400-6499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Fri Feb 19 14:26:17 2016		xmcs	
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XMCS: CODEC

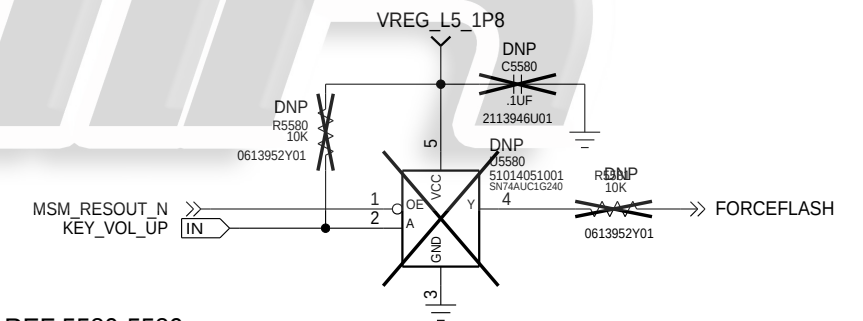
REF 2500-2579



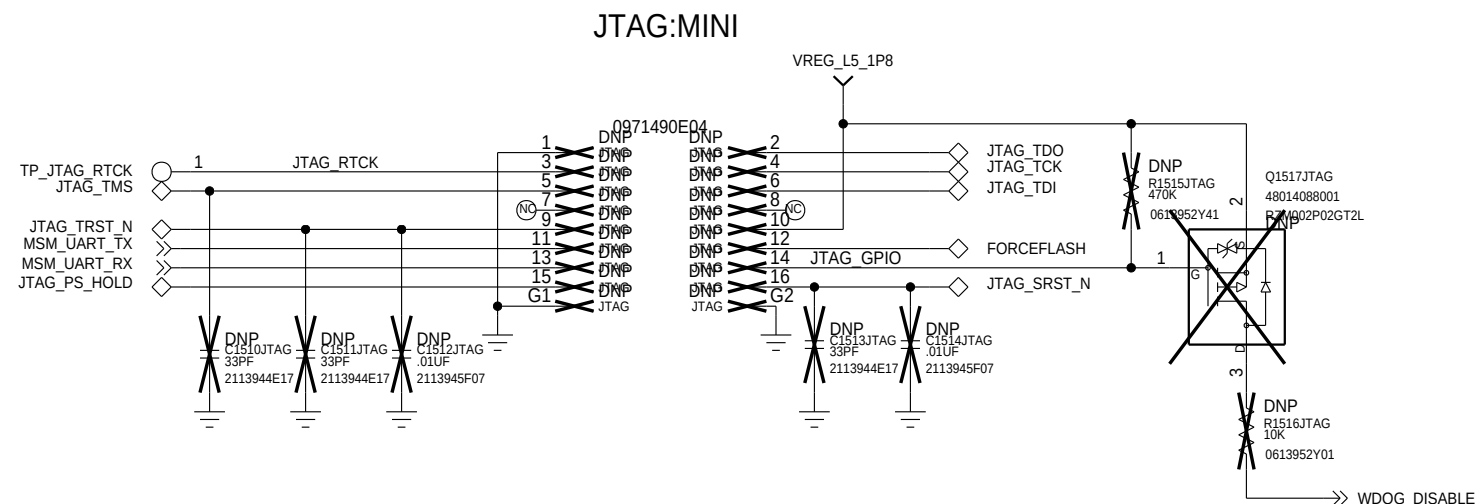
CAD NOTE:
PLACE C2532 AND C2533 CLOSE TO PIN A11 AND B11
PLACE C2534 CLOSE TO PIN G14
PLACE C2532 AND C2533 CLOSE TO PIN A11 AND B11
PLACE C2534 CLOSE TO PIN G14

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:16 2016		xmcs	
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 20 OF 52	

REF 1000-1099



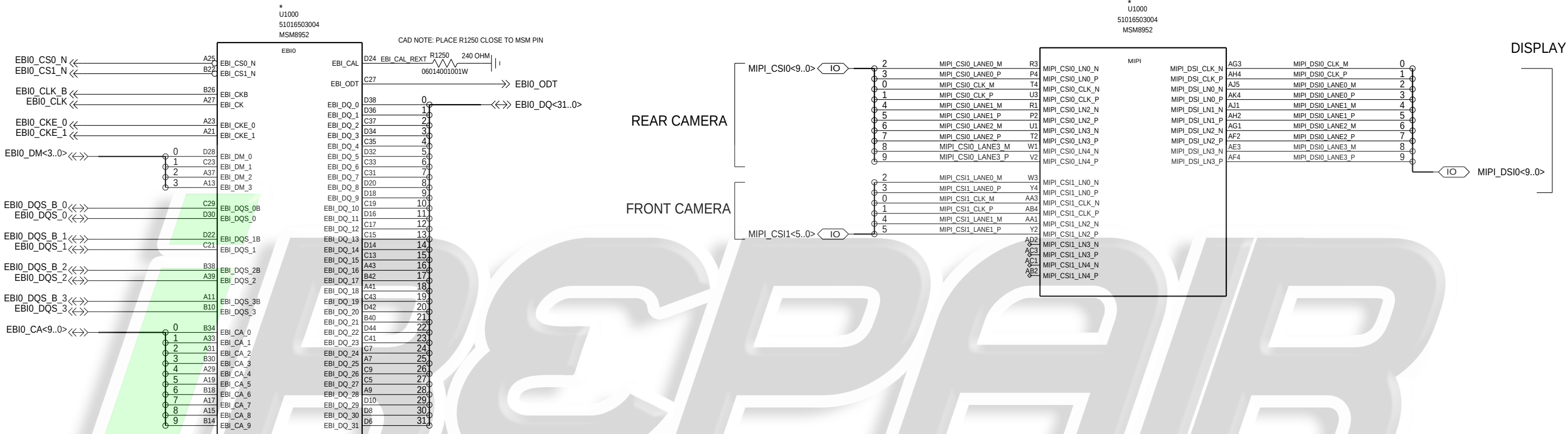
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:18 2016			
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		CAD FILE NAME:	PAGE 21 OF 52

MSM: MIPI/DSI/CSI/EBIO

REF 1200-1299



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		ap_pmic	
Fri Feb 19 14:26:26 2016			
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		PAGE 22 OF 52	

MSM: GPIO (0-133)

REF 1200-1299

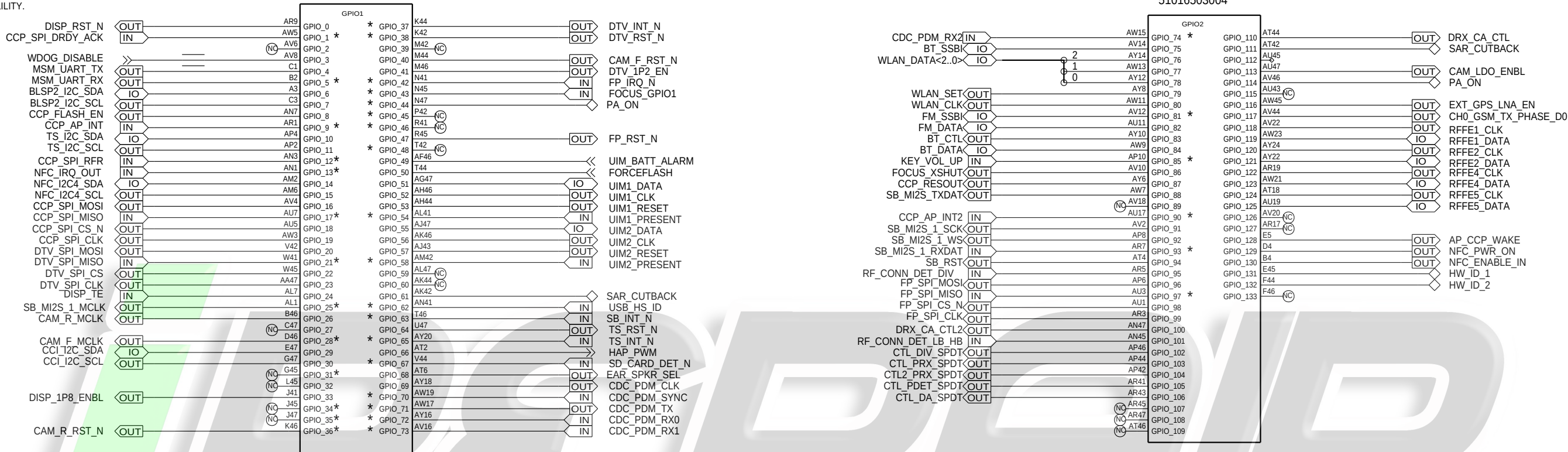
NOTE: RESERVE GPIO_3 FOR WDOG_DISABLE FUNCTION.
DO NOT HAVE A PULL-UP ON THIS GPIO FOR BETTER
SW DEBUG CAPABILITY.

U1000
MSM8952
51016503004

NOTE: ASTERISK(*) INDICATE MODEM POWER MANAGEMENT (MPM) WAKE-UP PINS.

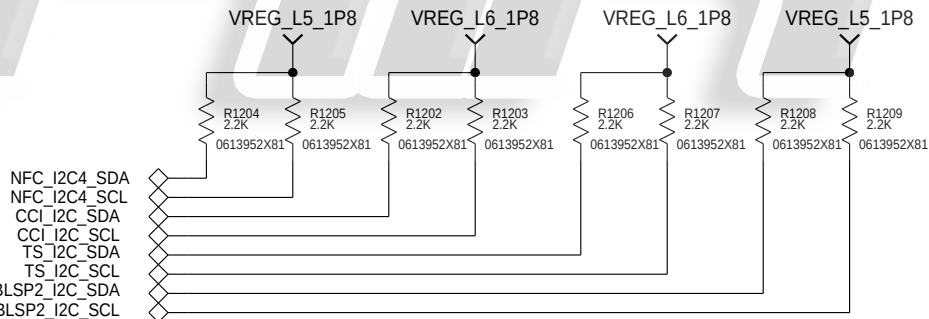
NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT
OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

U1000
MSM8952
51016503004



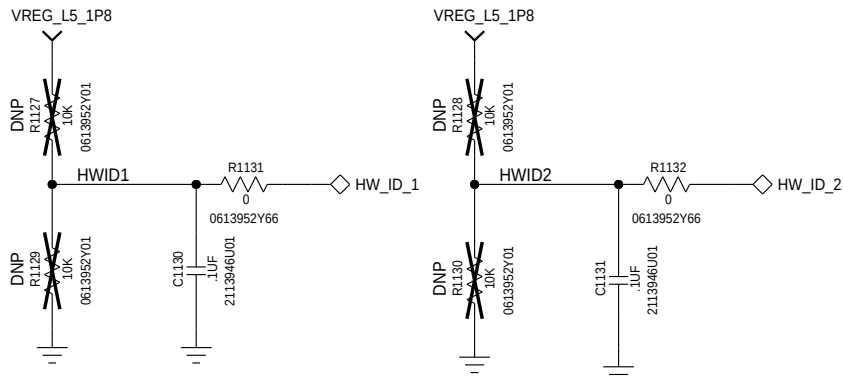
WAITING FOR GPIO ASSIGNMENT
NEED WAKABLE

I2C PULL-UP RESISTORS



MOT BOARD ID

FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>



APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Fri Feb 19 14:26:28 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME

84016530001_a.cpm

REV

01

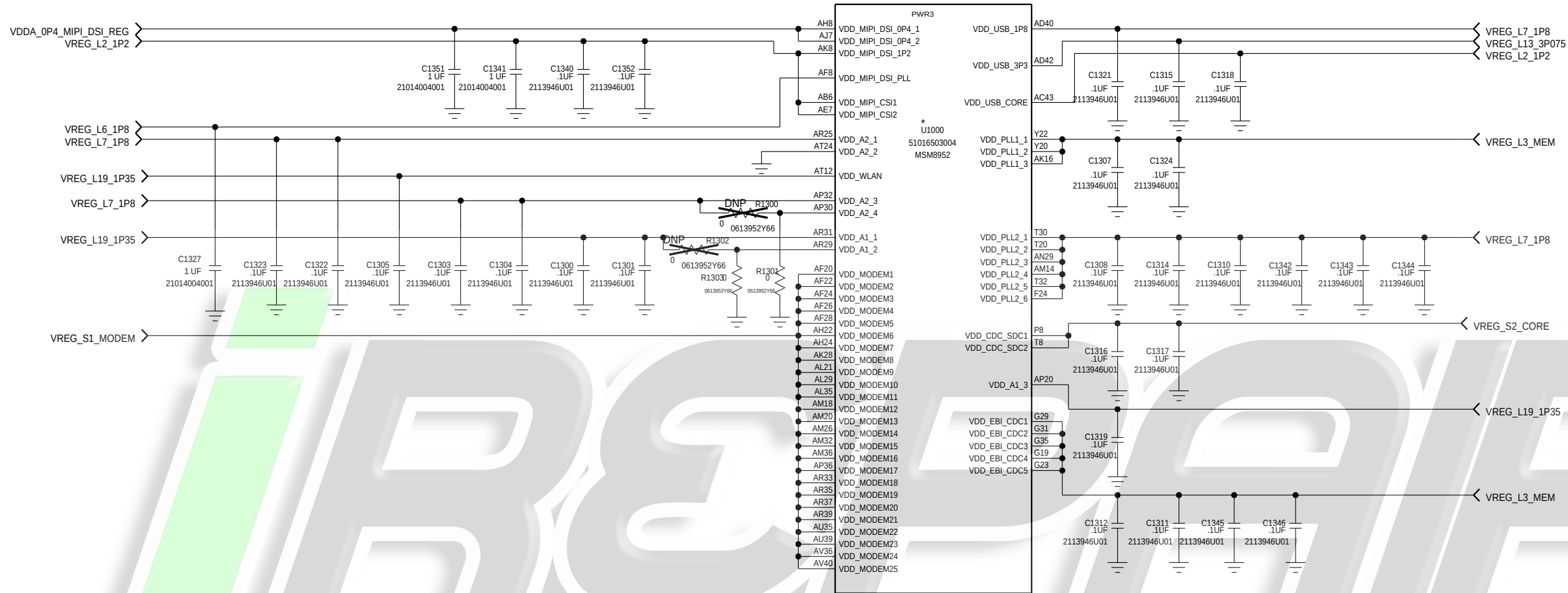
CAD FILE NAME:

PAGE 23 OF 52

MSM: PWR1

REF 1300-1399

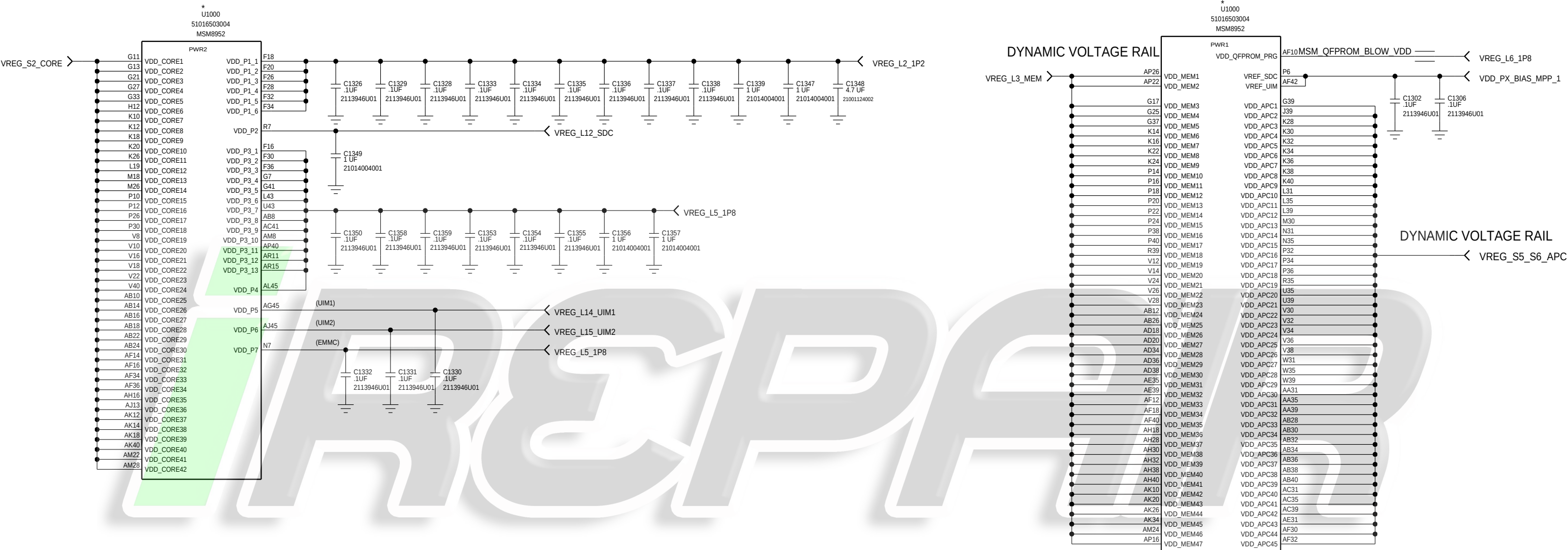
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:22 2016		ap_pmic	
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		CAD FILE NAME:	
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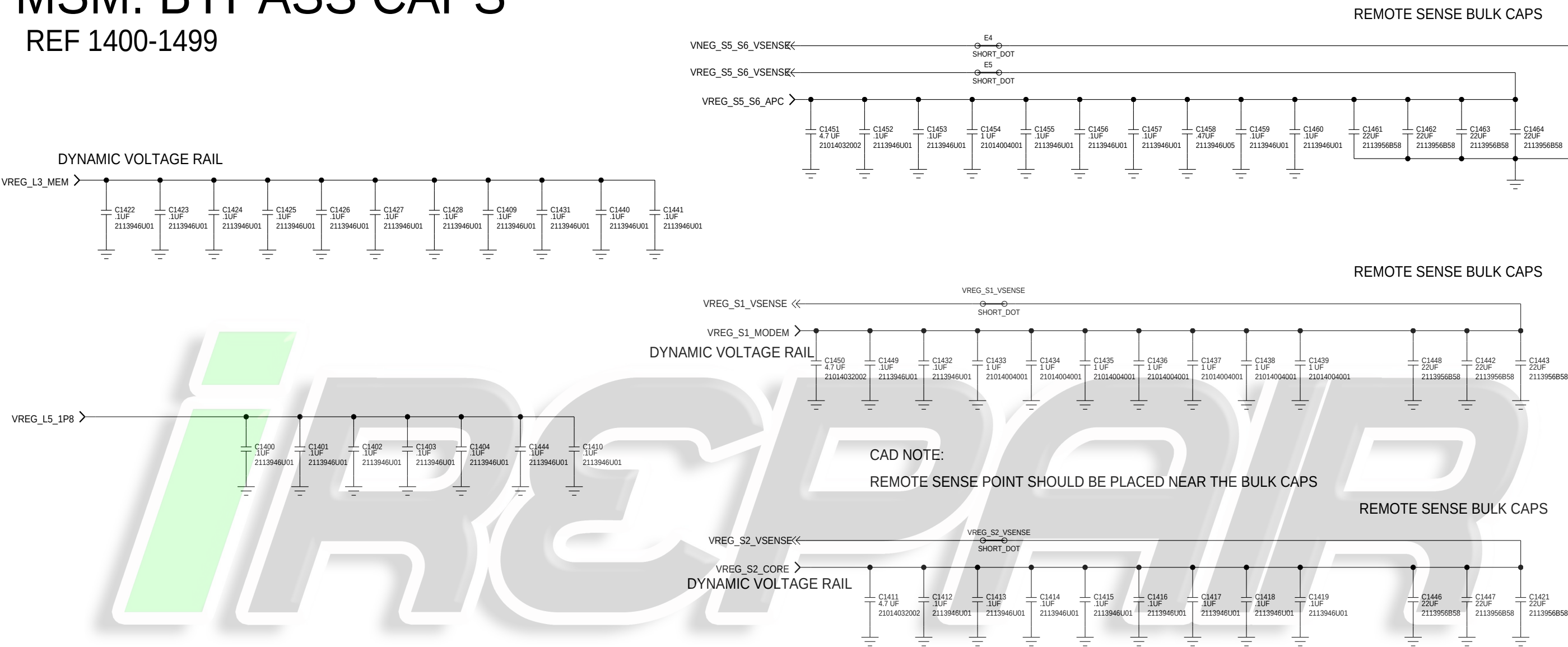
MSM: PWR2

REF 1300-1399



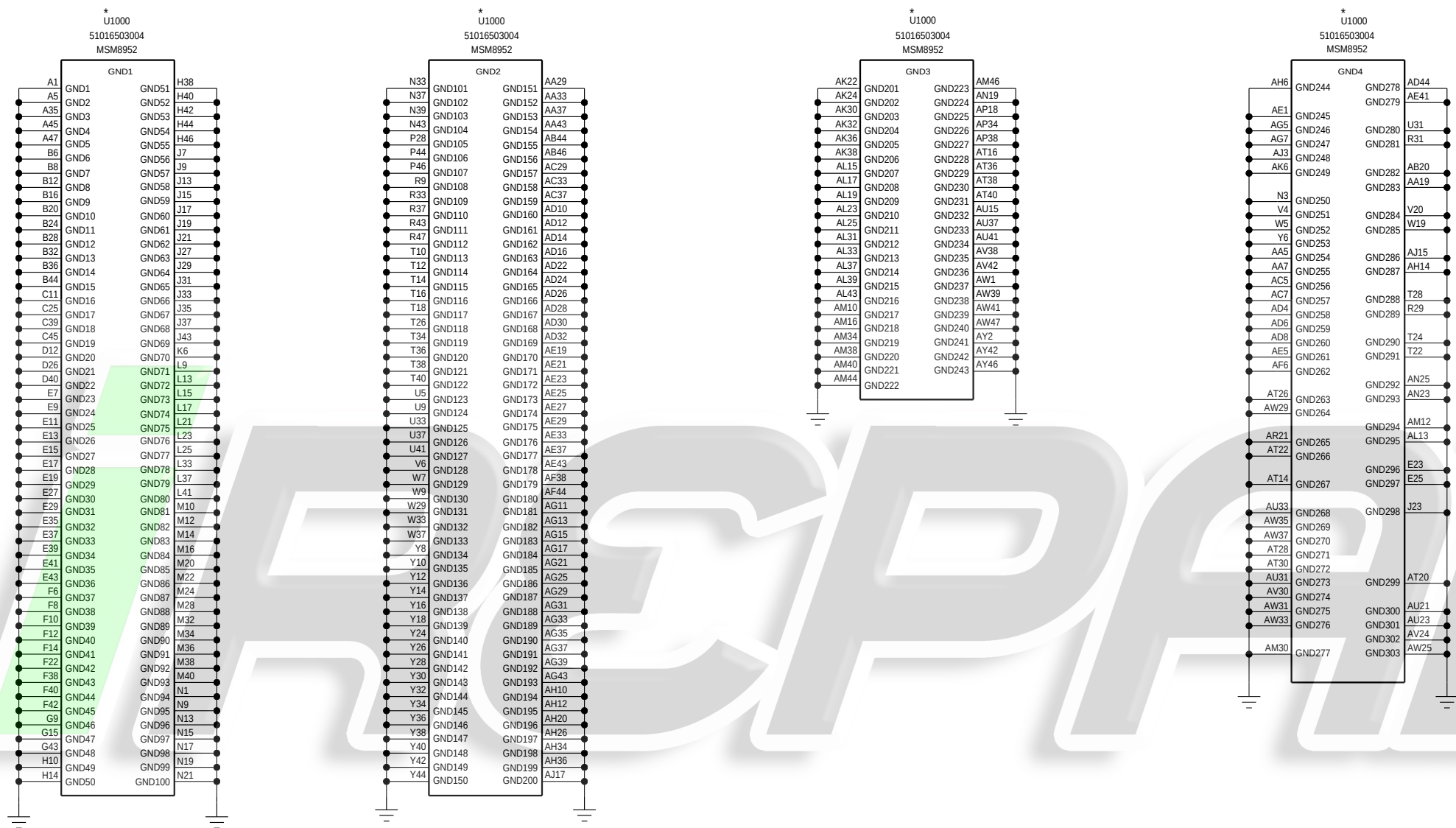
MSM: BYPASS CAPS

REF 1400-1499



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:24 2016		ap_pmic	
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MSM: GROUND



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 27 OF 52	

MSM: MCP MEMORY

MCP: 8GB NAND, 8GBIT LPDDR3

D

D

C

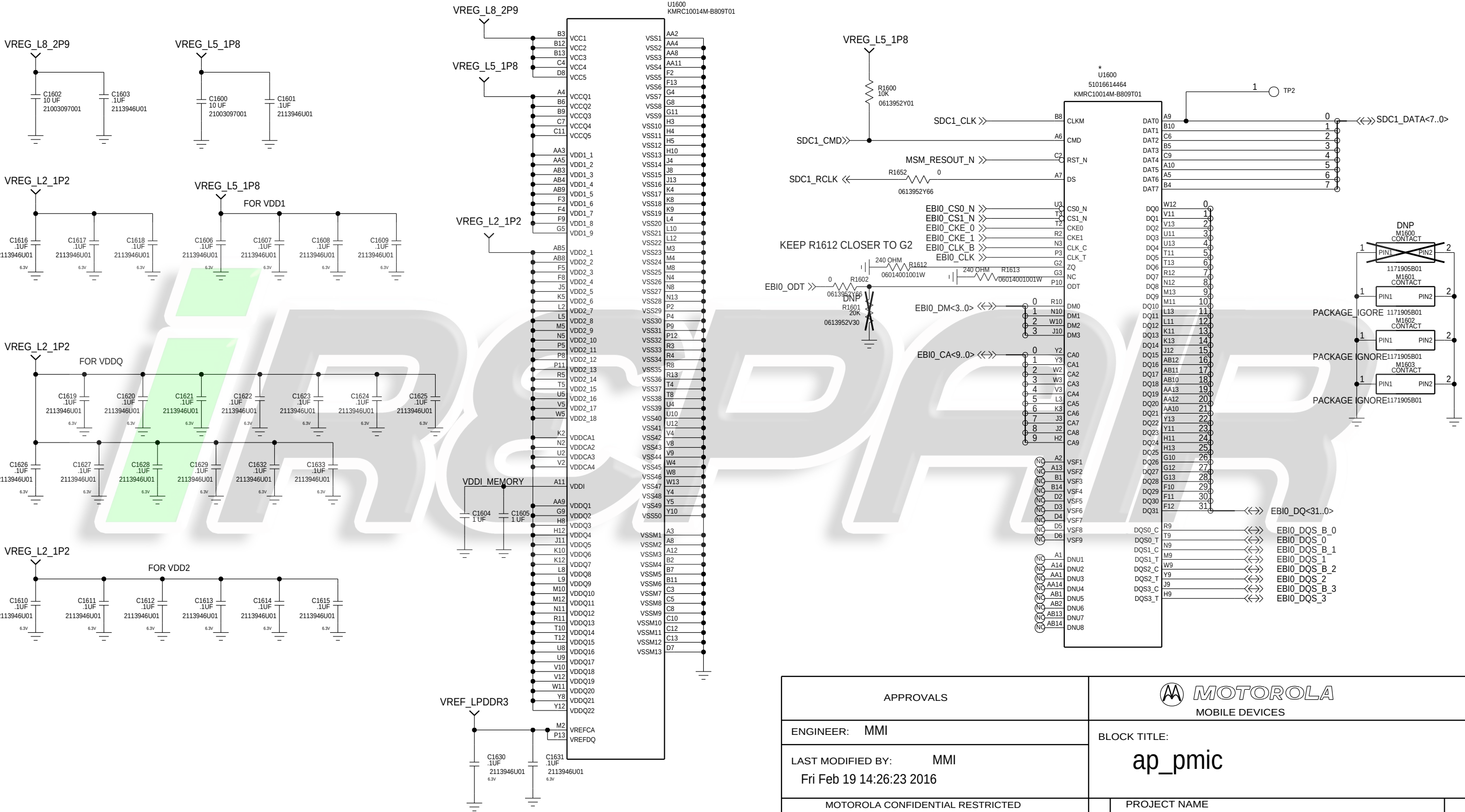
C

B

B

A

A



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:23 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 28 OF 52	

REF: 2000-2099

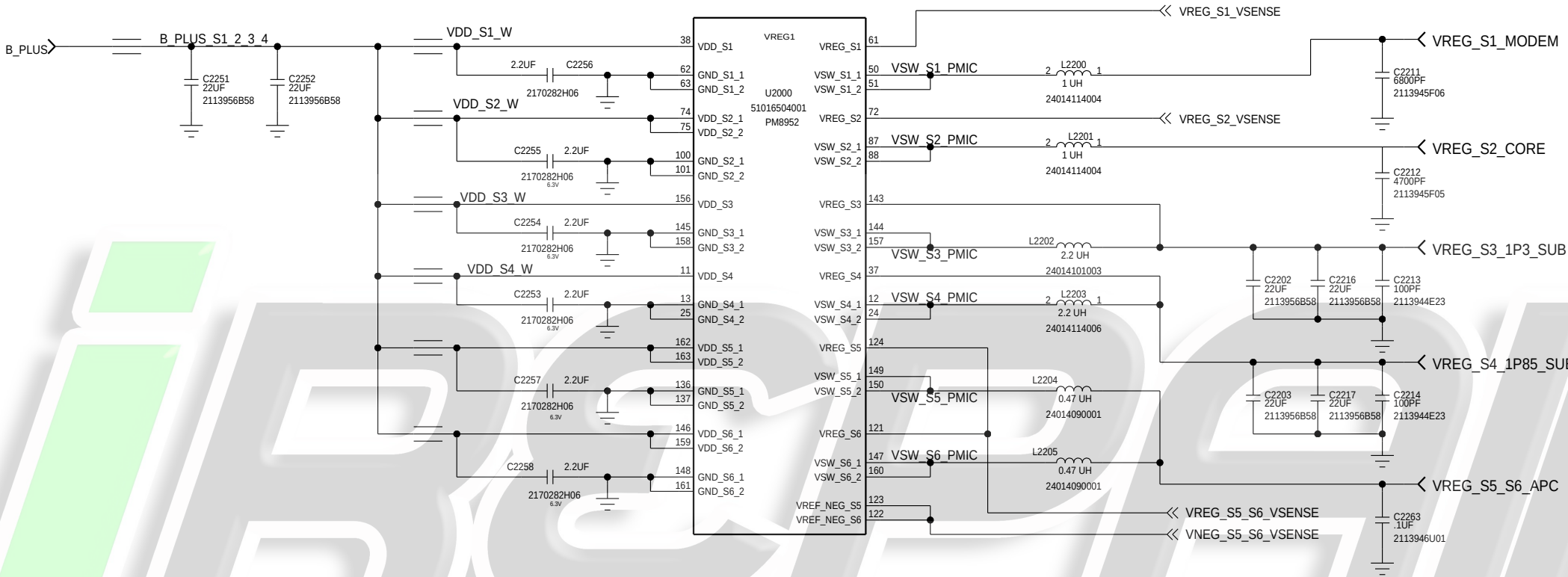


CAD NOTE 4: H3 MUST BE CONNECTED TO MAIN GND USING DEDICATED VIA

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:20 2016			
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		CAD FILE NAME:	PAGE 29 OF 52

PM8952: SMPS

REF: 2200-2299

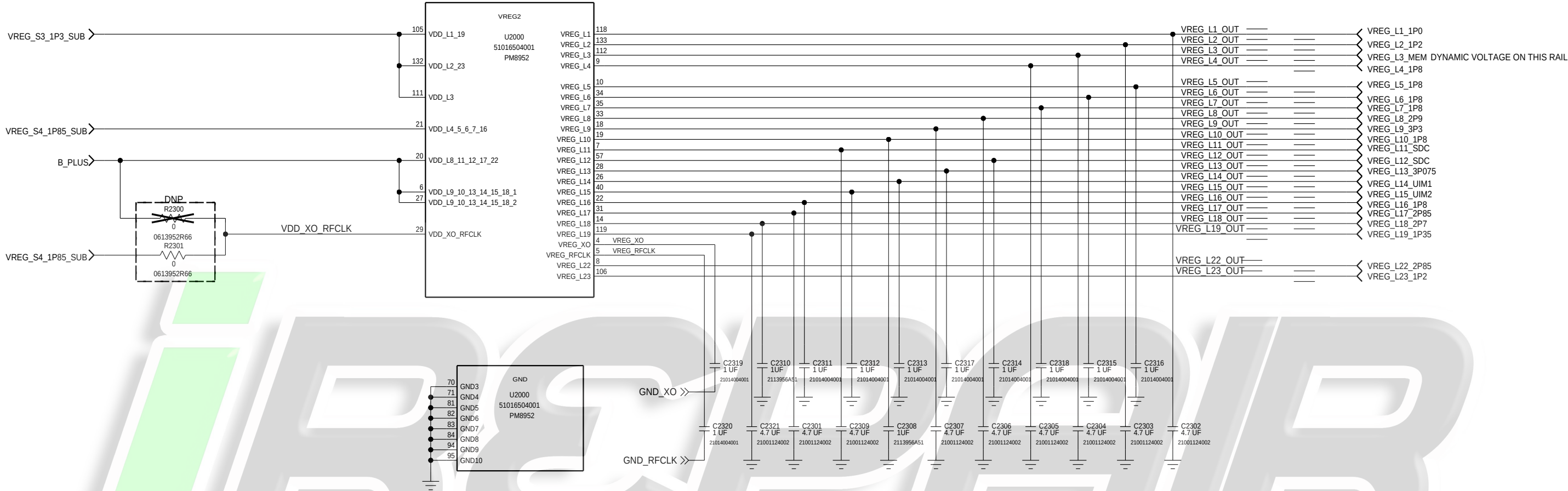


NOTE: PLACE INPUT CAPS CLOSE TO RESPECTIVE VIN PINS
EACH CAP GND SHOULD BE STAR-ROUTED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:25 2016		ap_pmic	
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 30 OF 52	

PM8952: VREGS

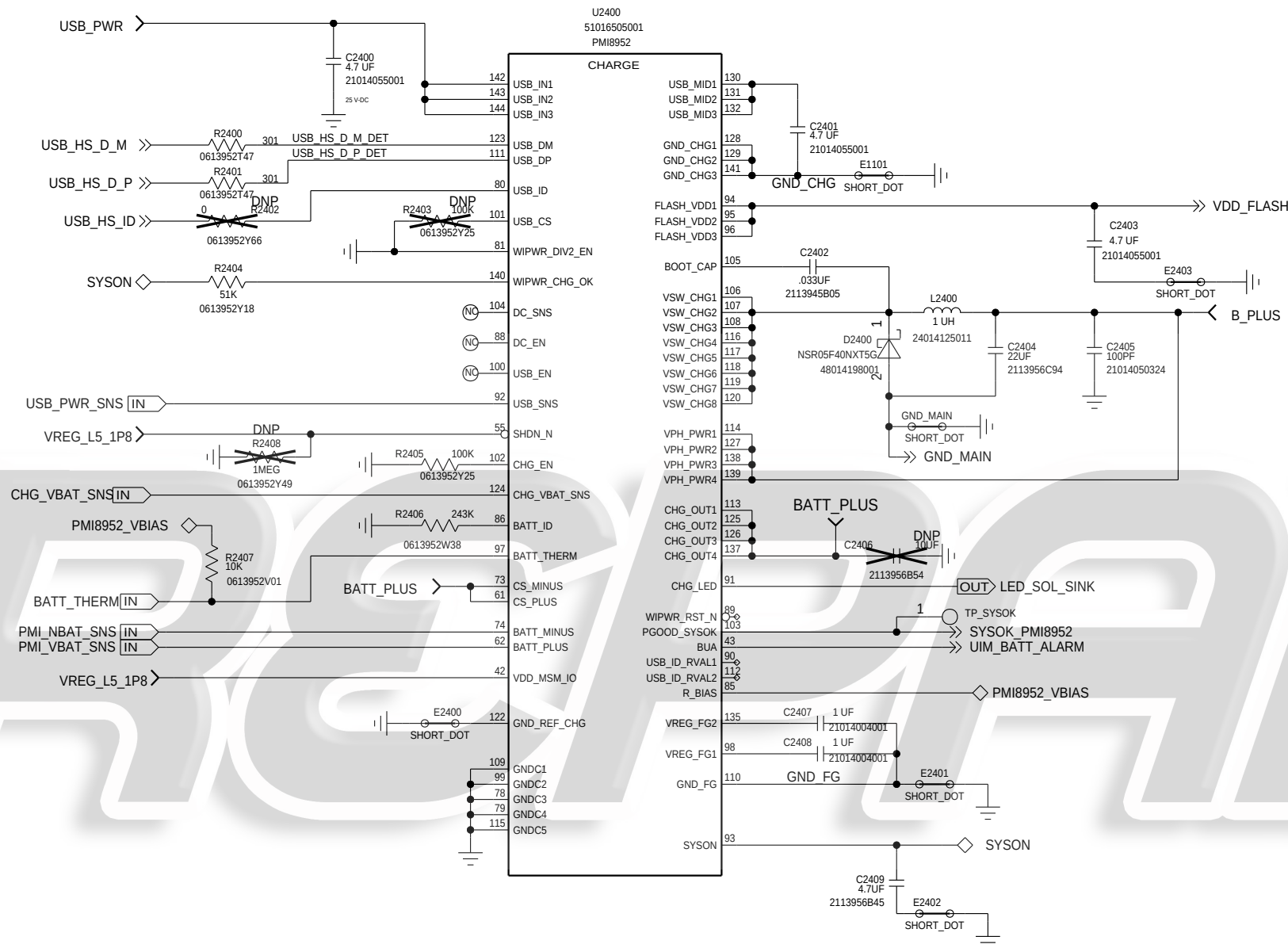
REF: 2300-2399



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		B 84016530001_a.cpm	
		CAD FILE NAME:	
		PAGE 31 OF 52	

PMI8952: CHARGING

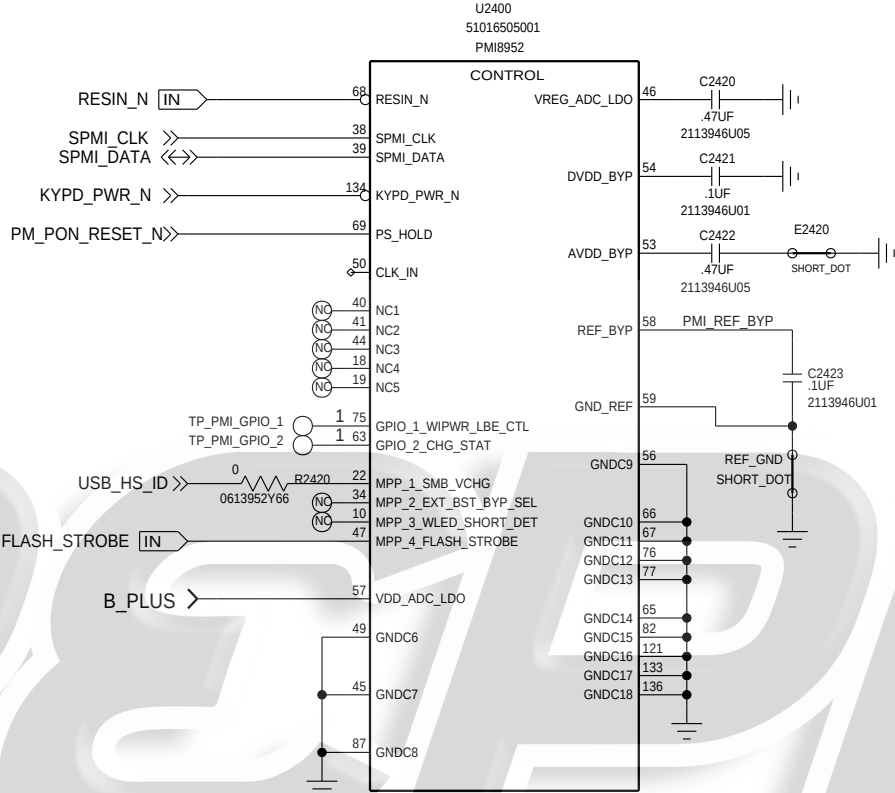
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:19 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 32 OF 52	

PMI8952: IO_REF

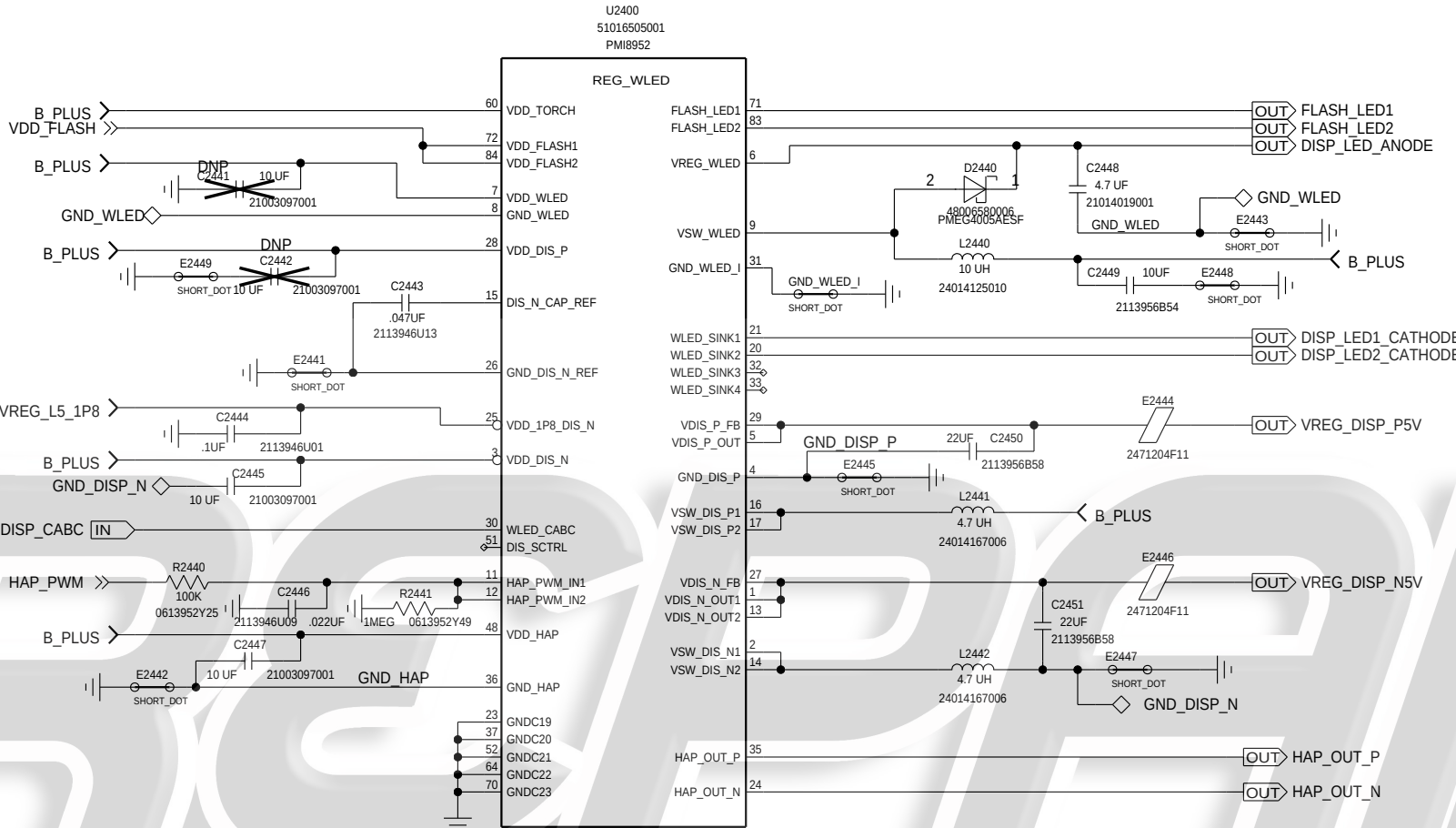
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		B 84016530001_a.cpm	01
		CAD FILE NAME:	
		PAGE 33 OF 52	

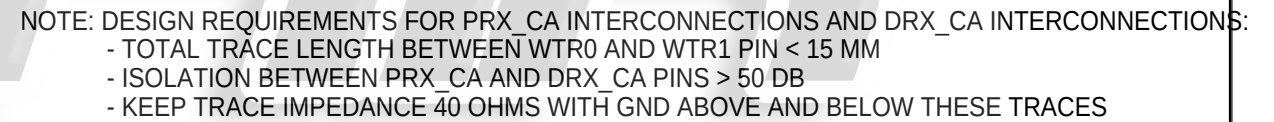
PMI8952: REG_WLED

REF: 2440-2459



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:21 2016		ap_pmic	
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
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REF 3000-3049



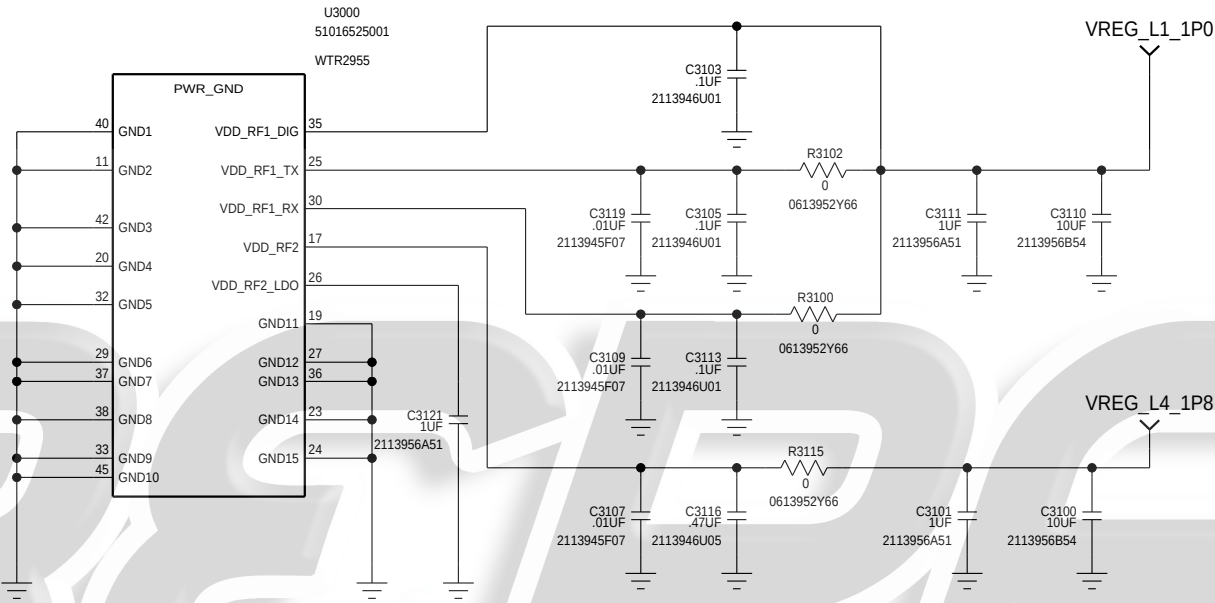
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Fri Feb 19 14:26:30 2016			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME B 84016530001_a.cpm	REV 01
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BASED ON 80-NP237-43 REV. A

WTR2955 POWER

REF 3100-3149



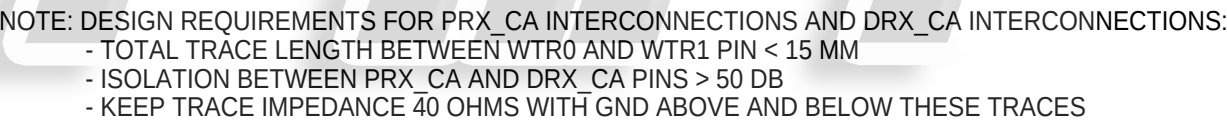
NOTE: C3119, C3109, C3107 ARE PRIORITY 1 AND SHOULD BE PLACED CLOSEST TO VDD PINS.

BASED ON 80-NL713-44 REV. A

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

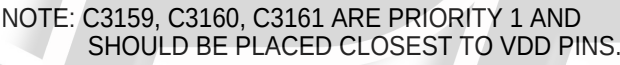
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ENGINEER: MMI		BLOCK TITLE:	
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		CAD FILE NAME:	
		PAGE 36 OF 52	

REF 3050-3099



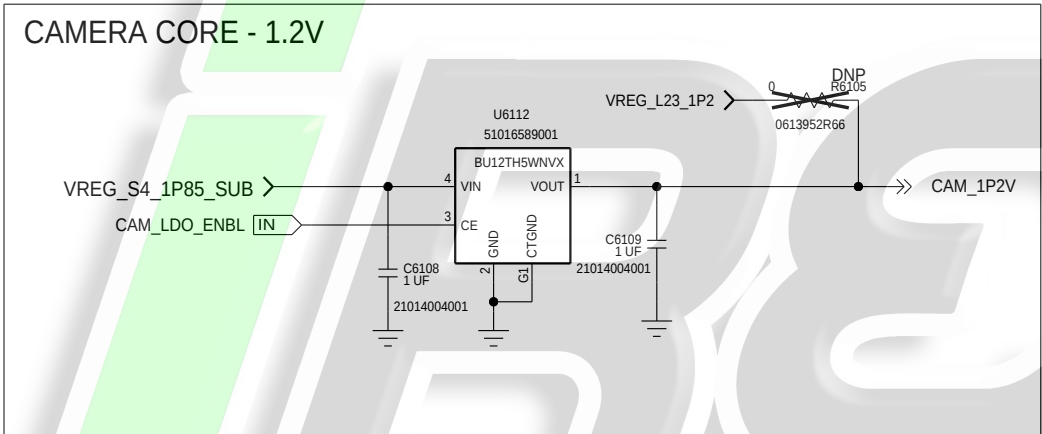
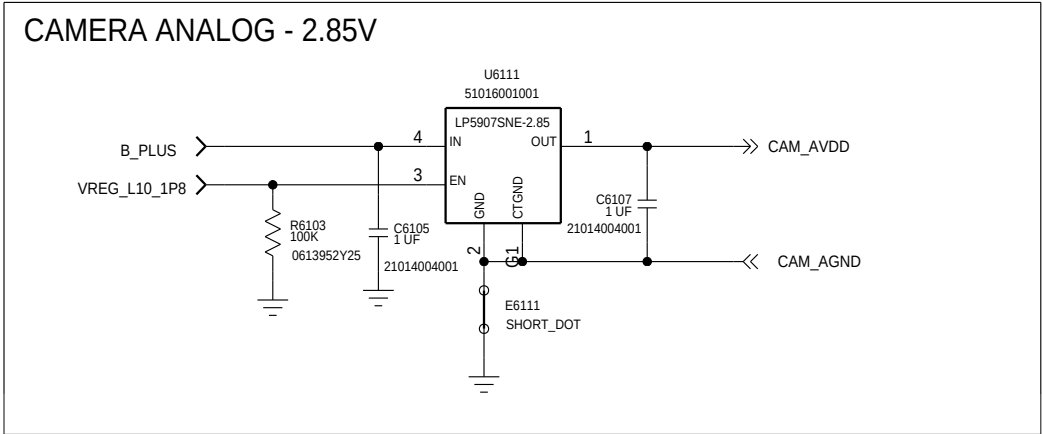
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ENGINEER: MMI	BLOCK TITLE: ca_wtr2955		
LAST MODIFIED BY: Fri Feb 19 14:26:32 2016			
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		CAD FILE NAME:	PAGE 37 OF 52

REF 3150-3199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: ca_wtr2955	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:32 2016			
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		CAD FILE NAME:	PAGE 38 OF 52

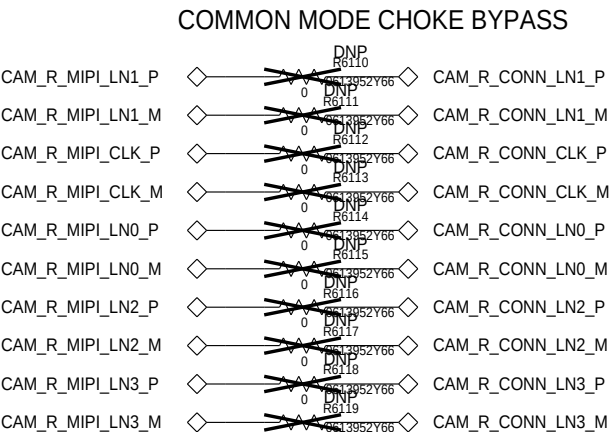
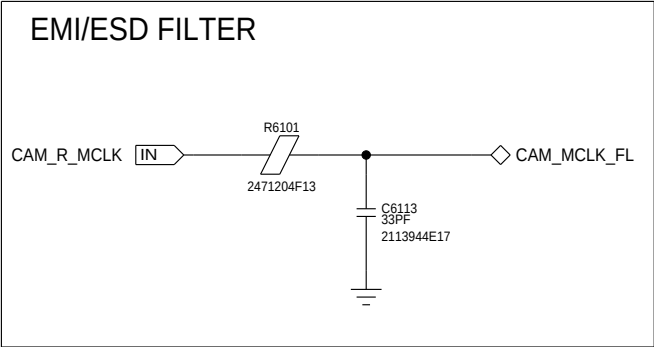
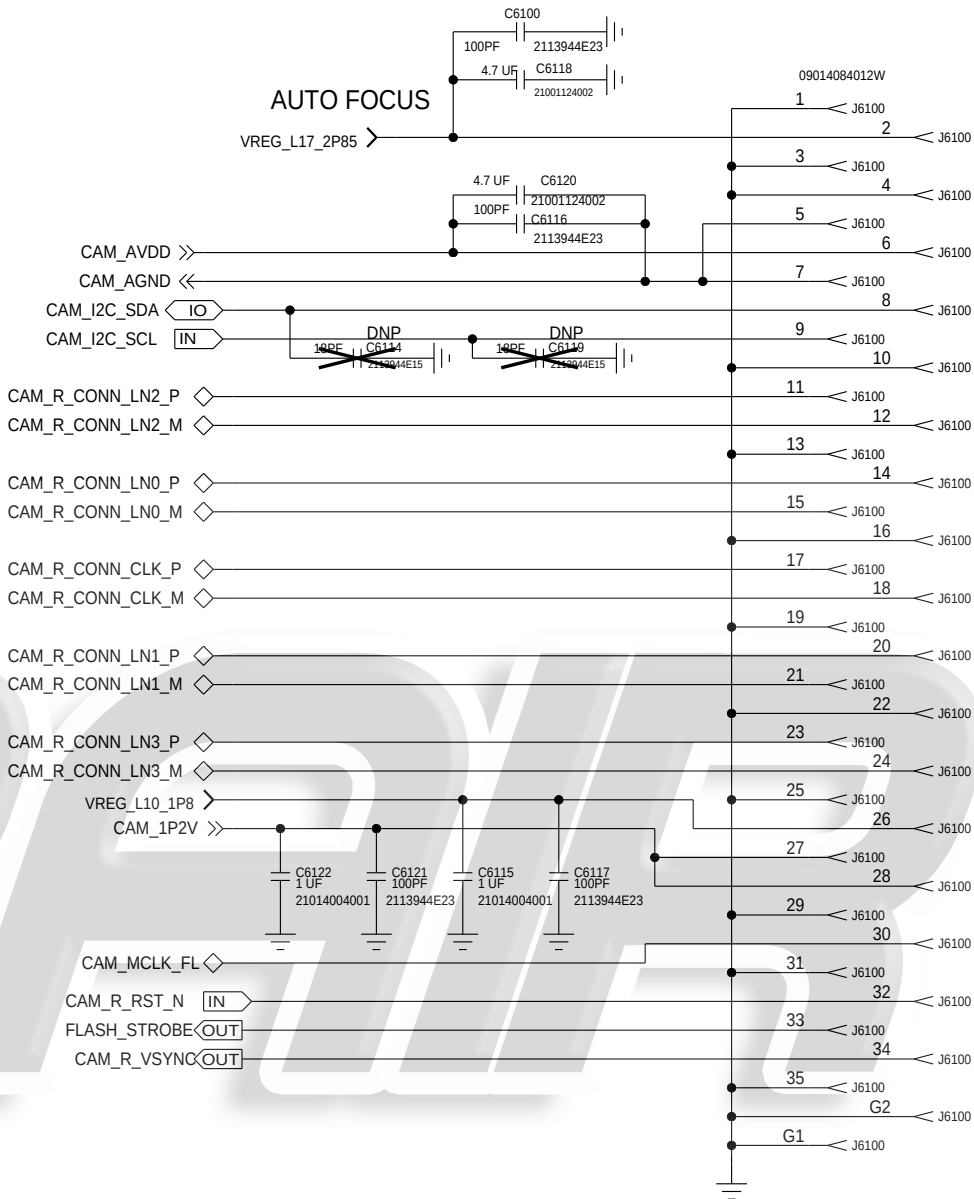
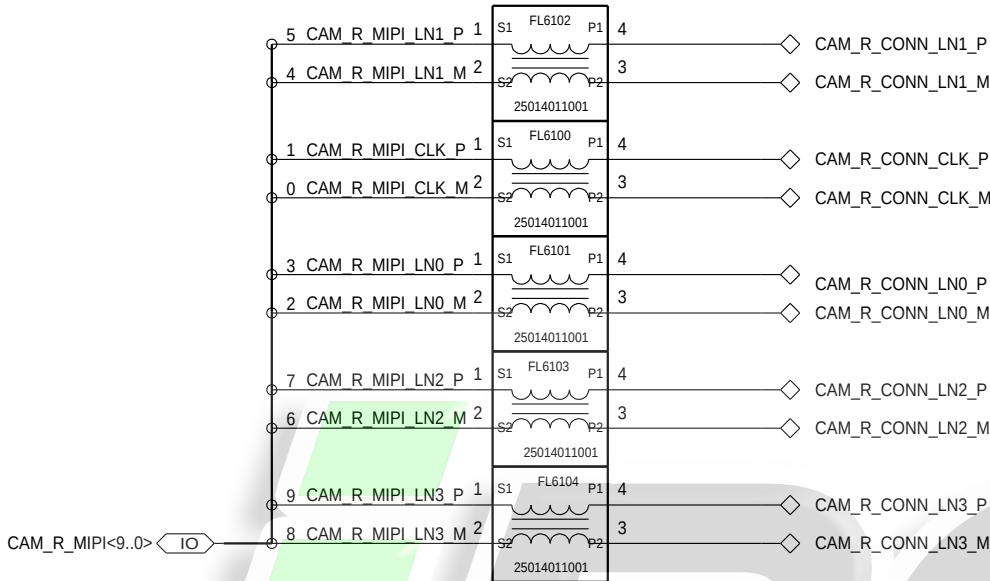
CAM: POWER



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:33 2016		multimedia	
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		CAD FILE NAME:	
		PAGE 39 OF 52	

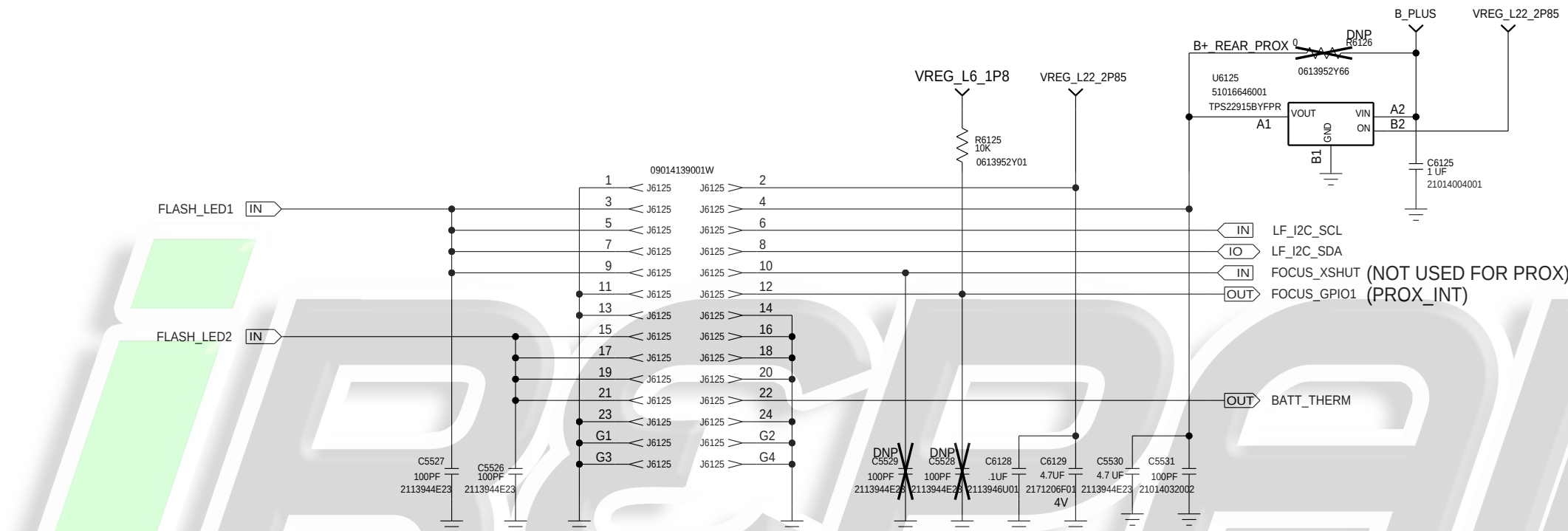
CAM: REAR

REF:6100-6124



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE: multimedia	
LAST MODIFIED BY: AZIZ NOORI Fri Feb 19 14:26:34 2016		PROJECT NAME B 84016530001_a.cpm	
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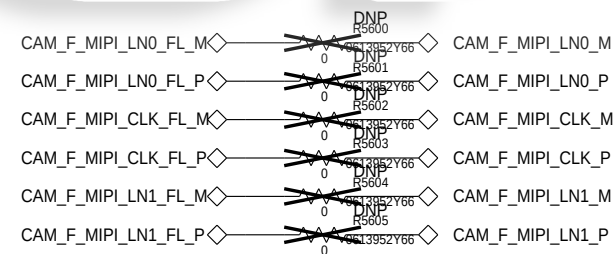
LASER FOCUS & FLASH CONN



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH WOLNY		BLOCK TITLE: multimedia	
LAST MODIFIED BY: HUNG PHAM Fri Feb 19 14:26:36 2016		PROJECT NAME B 84016530001_a.cpm	
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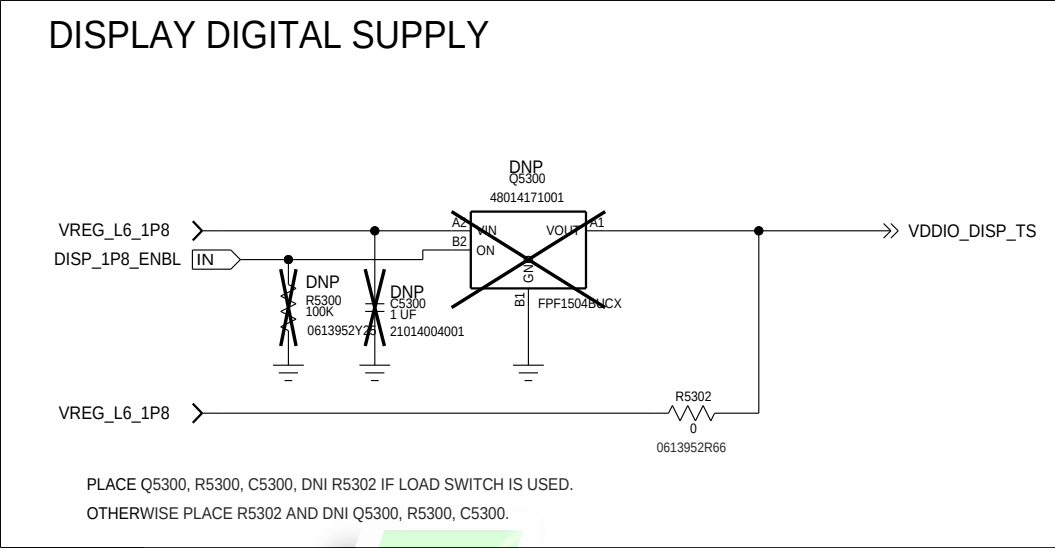
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FRONT CAM I2C ADDR:



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER	BLOCK TITLE: multimedia		
LAST MODIFIED BY: NATE CONNELL Tue Feb 16 11:14:52 2016			
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		CAD FILE NAME:	PAGE 42 OF 52

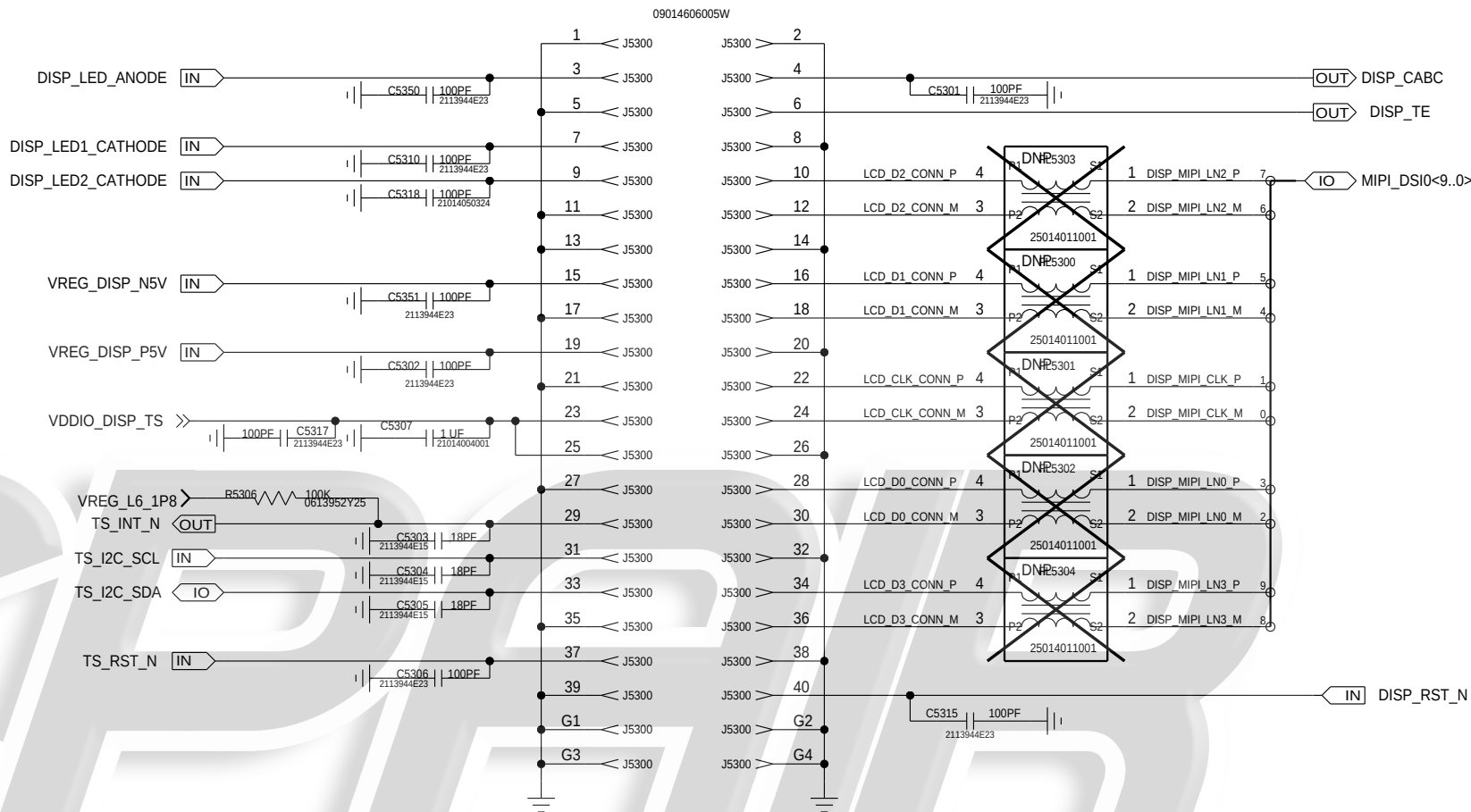
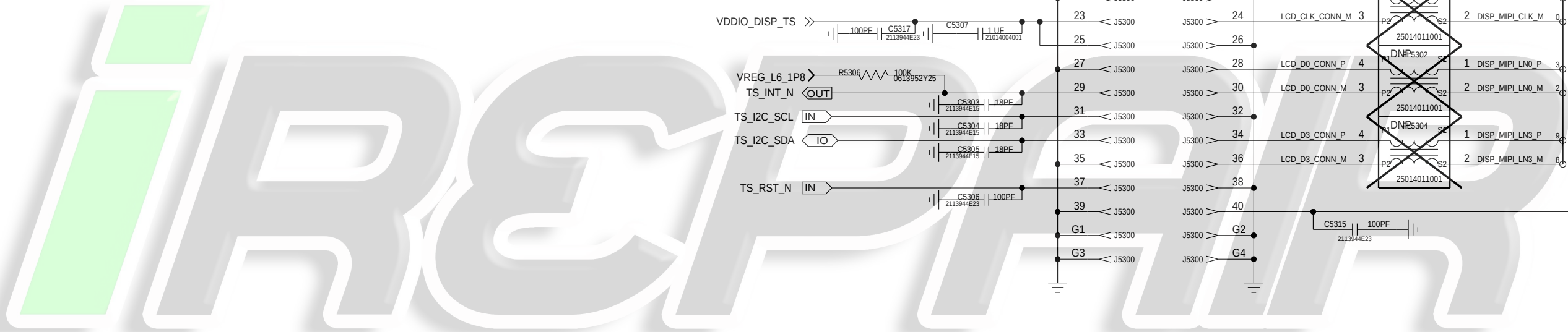
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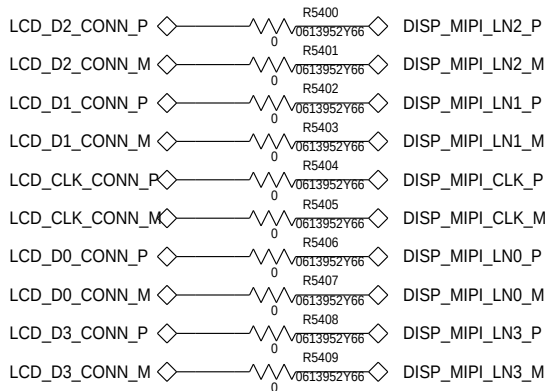
REPAIR

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:36 2016			
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		CAD FILE NAME:	PAGE 43 OF 52

DISP: CONNECTOR

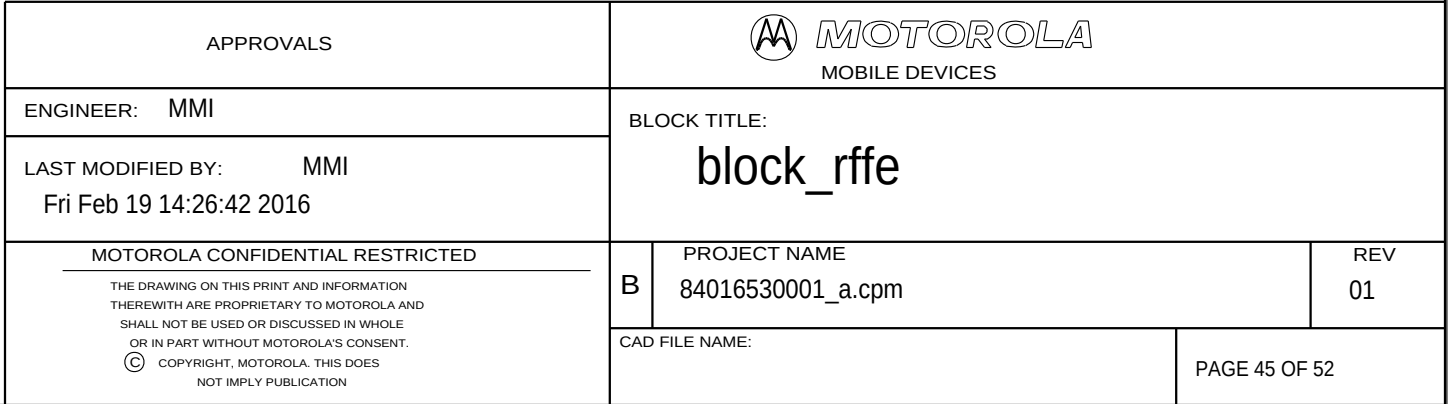


COMMON MODE CHOKE BYPASS



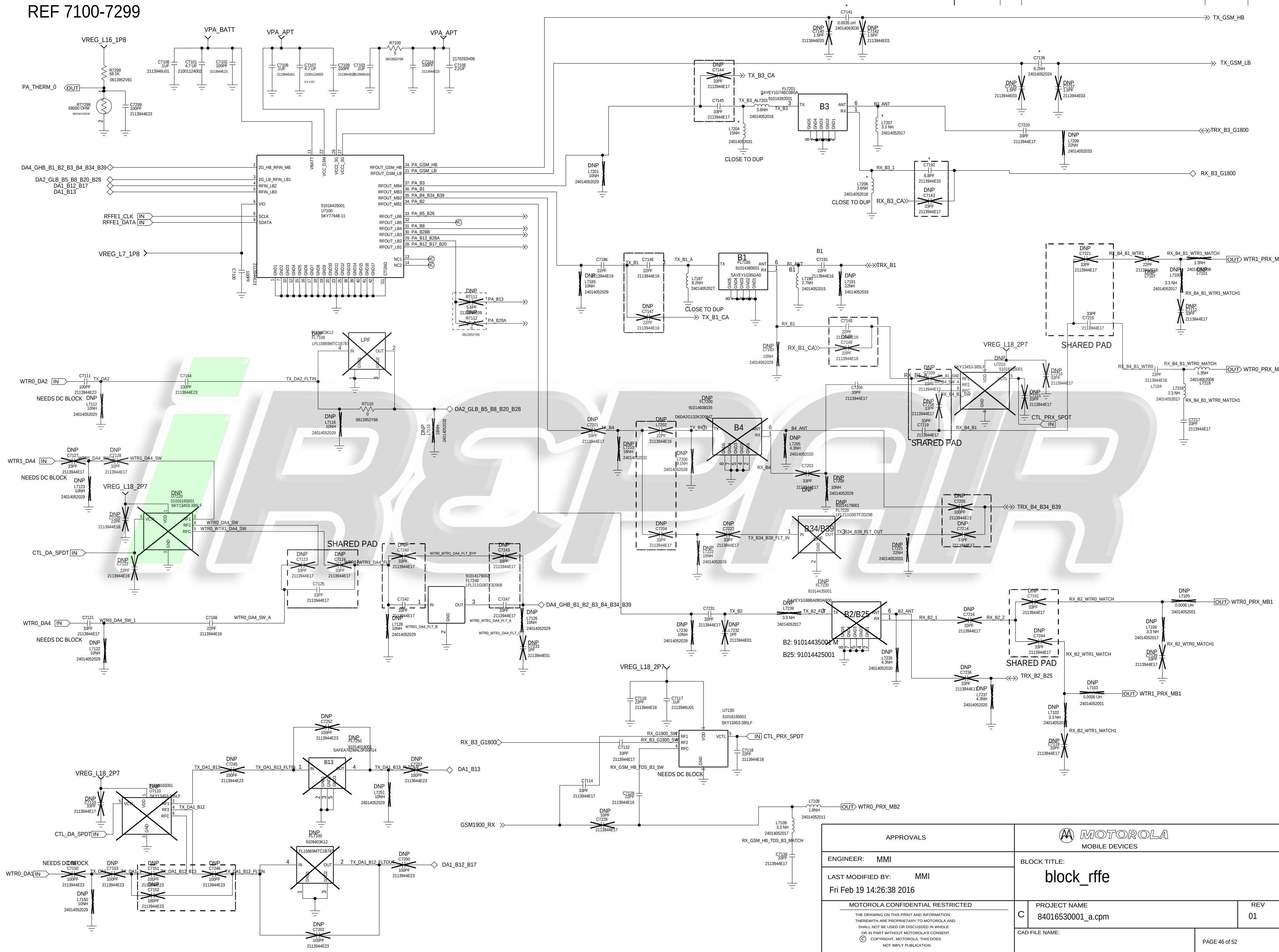
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REF 7000-7099



REF 7100-7299

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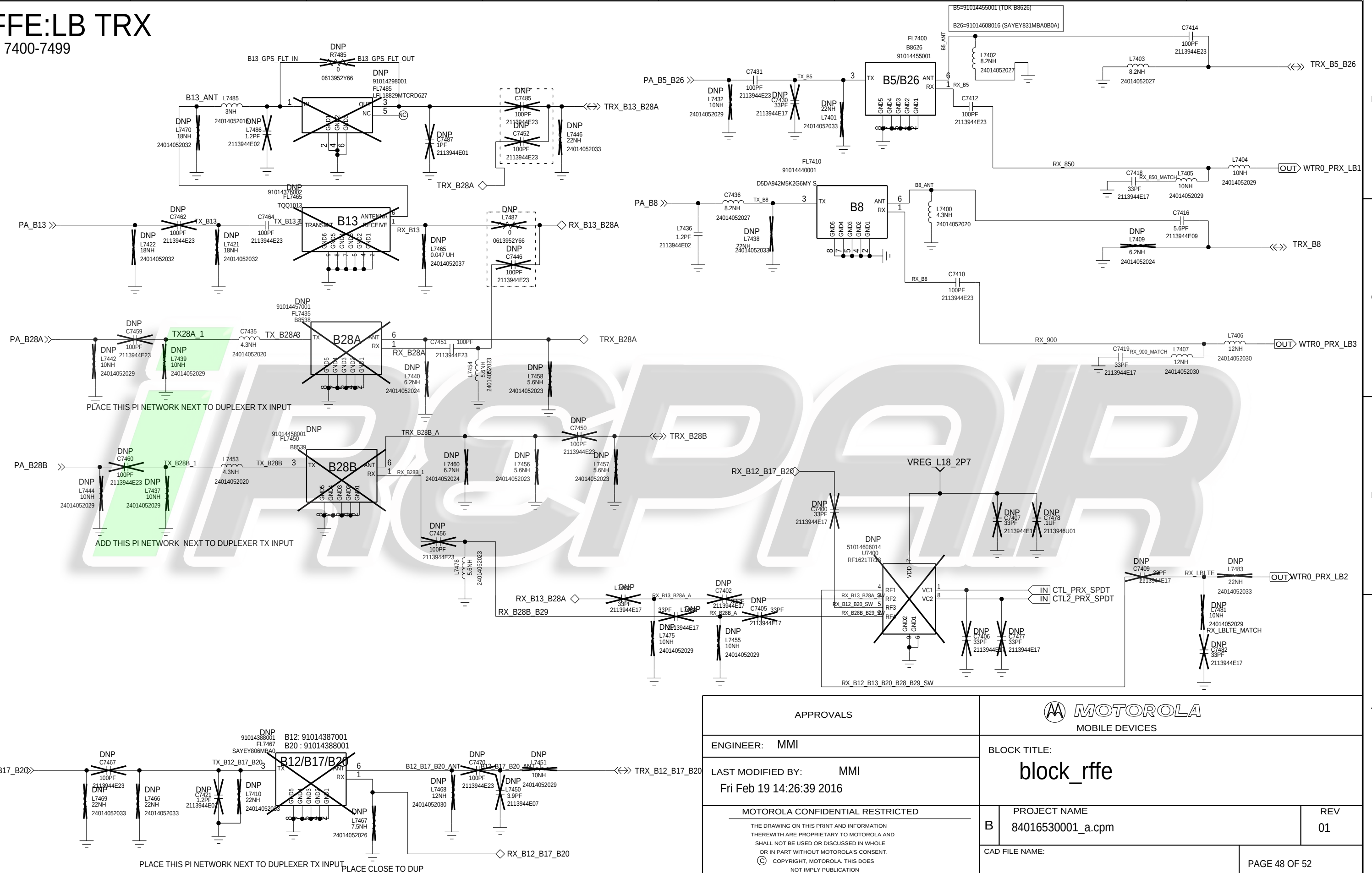
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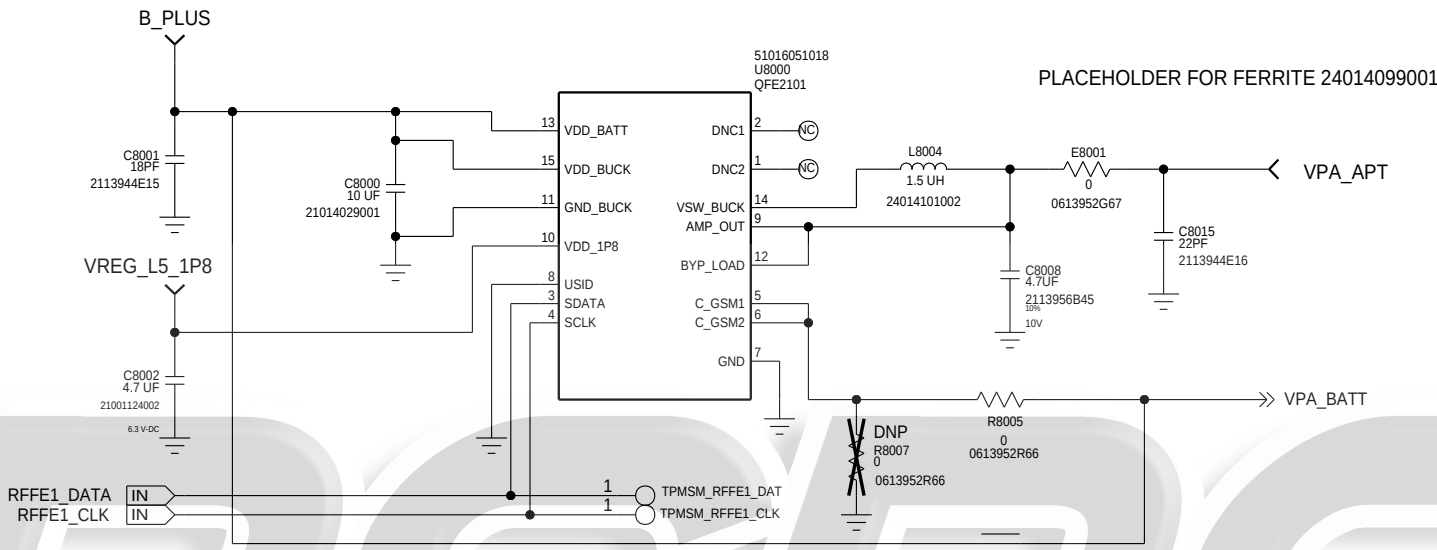
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:40 2016				
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		CAD FILE NAME:		PAGE 47 OF 52

RFFE:LB TRX
REF 7400-7499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: block_rffe		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:39 2016			
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		CAD FILE NAME:	PAGE 48 OF 52

RFFE:QFE2101
REF 7400-7499



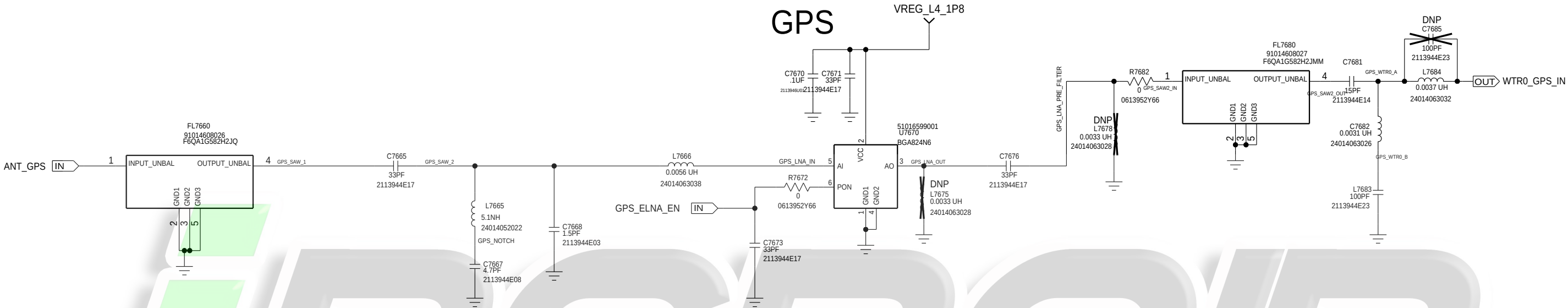
GND_BUCK: DO NOT VIA DIRECTLY TO GROUND PLANE. KEEP ISOLATED ON SURFACE LAYER;
ROUTE TO INPUT CAPACITOR GROUND PAD, AND THEN ONTO A VIA THAT CONNECTS TO THE GROUND
PLANE AT LEAST 1 MM BEYOND THE CAPACITOR.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: block_rffe	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:37 2016			
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		CAD FILE NAME: PAGE 49 OF 52	

VREG L18 2P7

RFFE: GPS

REF 7650-7699



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
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		PROJECT NAME	REV
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		CAD FILE NAME:	
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